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(54) **FIBROUS STRUCTURE-CONTAINING ARTICLES**

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D21H 27/00 (2006.01)
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(58) **Field of Classification Search**

None

See application file for complete search history.

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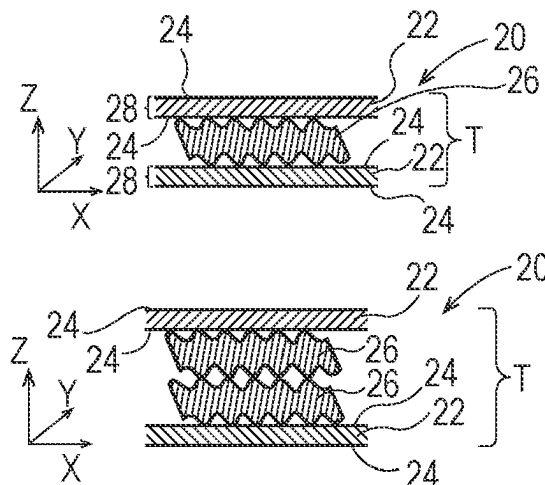
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(57) **ABSTRACT**

Articles, such as sanitary tissue products, including fibrous structures, and more particularly articles including fibrous structures having a plurality of fibrous elements wherein the article exhibits differential cellulose content throughout the thickness of the article and methods for making same are provided.

6 Claims, 14 Drawing Sheets



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D01D 5/098 (2006.01)
D01D 5/18 (2006.01)
D21H 15/10 (2006.01)
D21H 21/20 (2006.01)
D21H 21/22 (2006.01)
D21H 21/50 (2006.01)
D21H 27/30 (2006.01)
D21H 27/32 (2006.01)
D21H 27/38 (2006.01)

(52) **U.S. Cl.**

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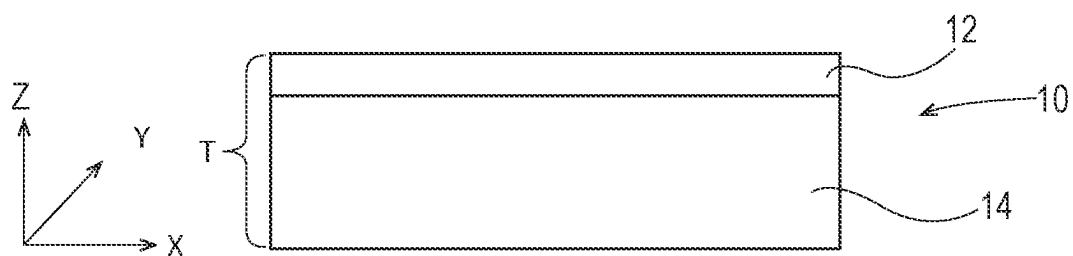


Fig. 1
PRIOR ART

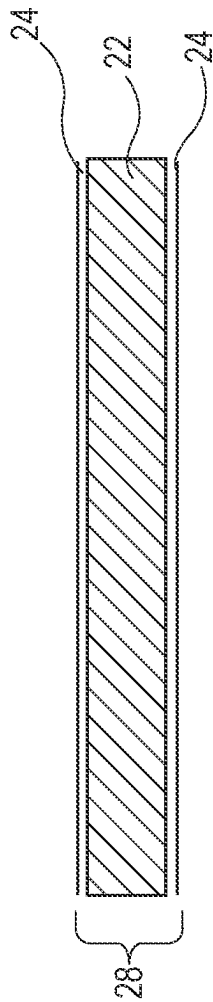
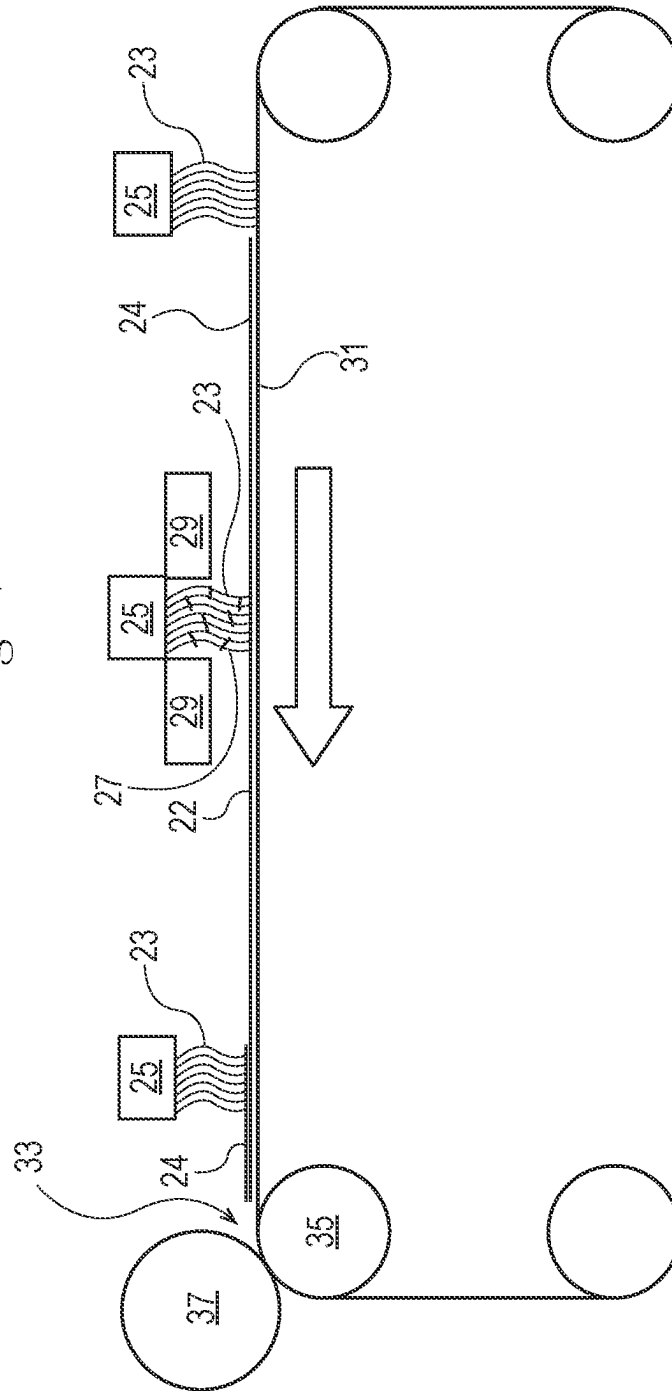


Fig. 2A



2B
Fis

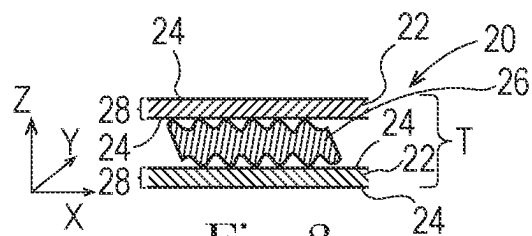


Fig. 3

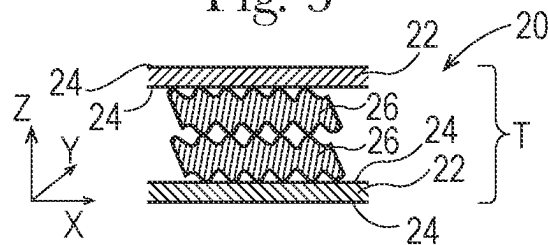


Fig. 4

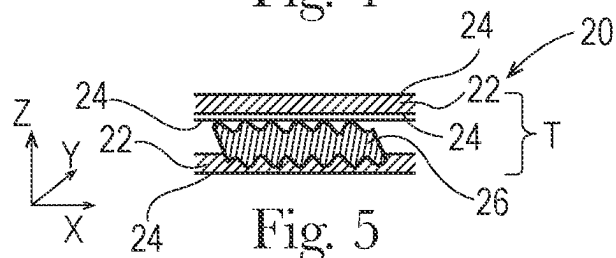


Fig. 5

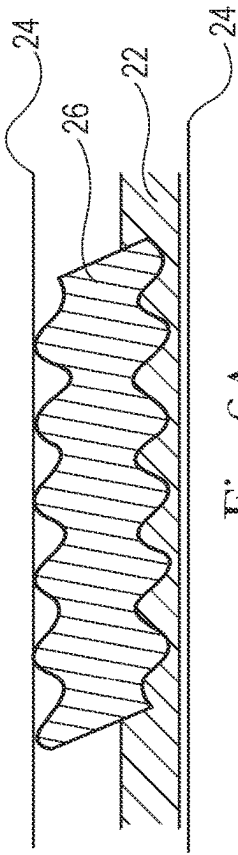


Fig. 6A

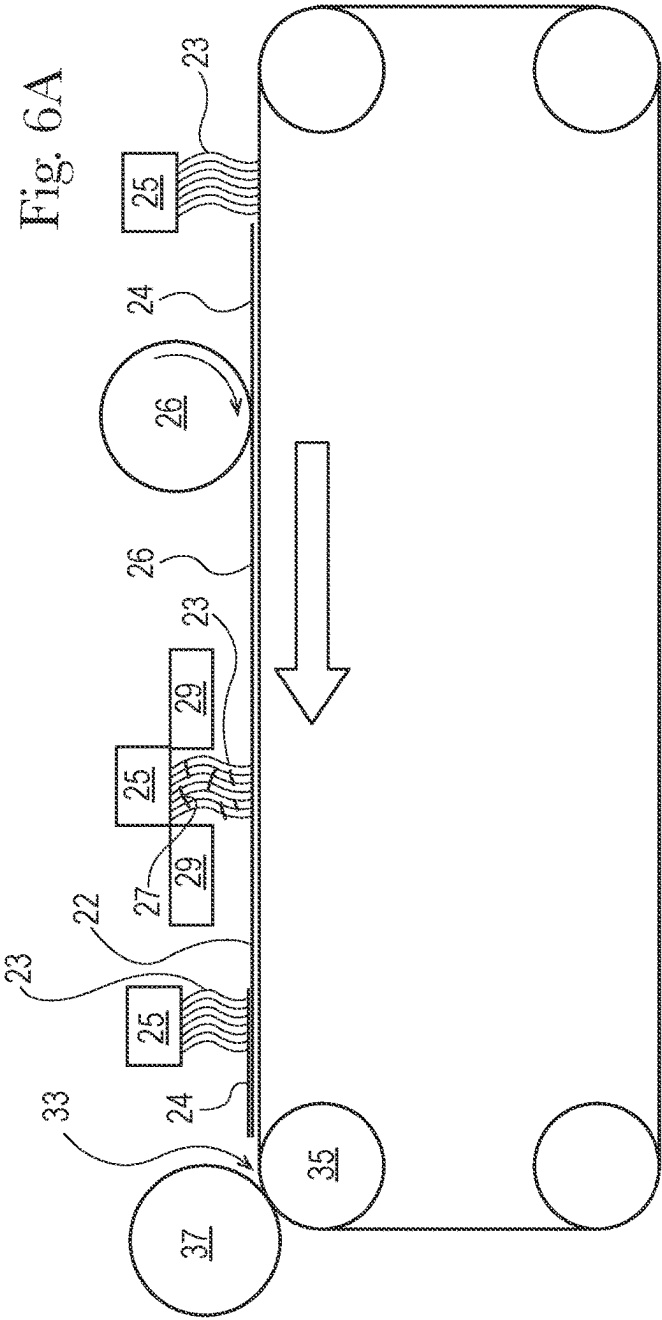


Fig. 6B

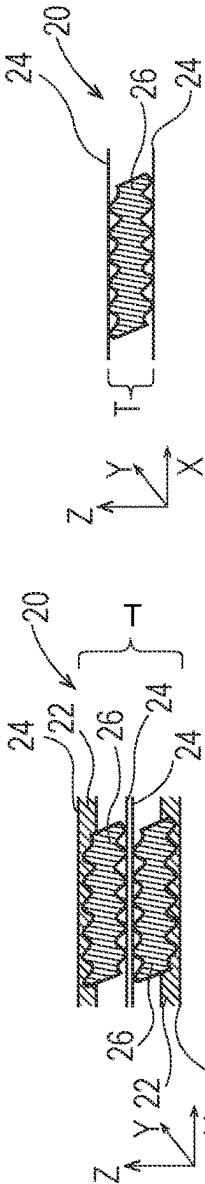


Fig. 7

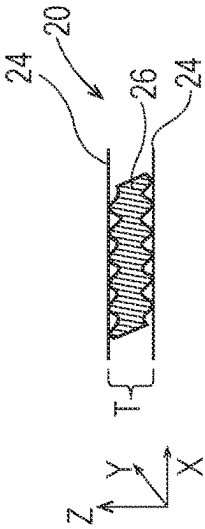


Fig. 8

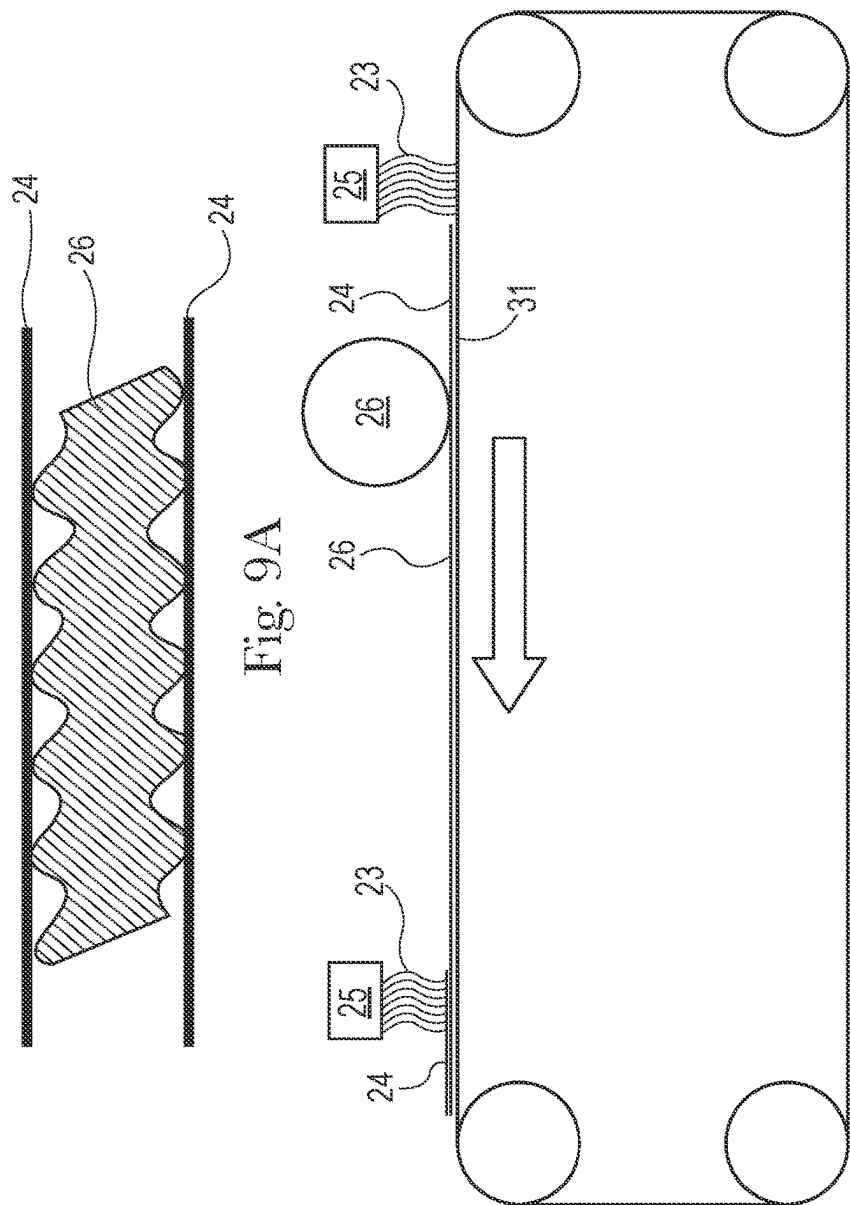


Fig. 9A

Fig. 9B

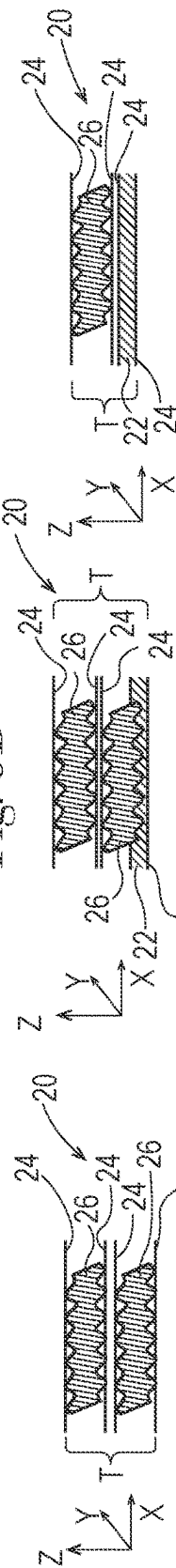


Fig. 12

Fig. 11

Fig. 10

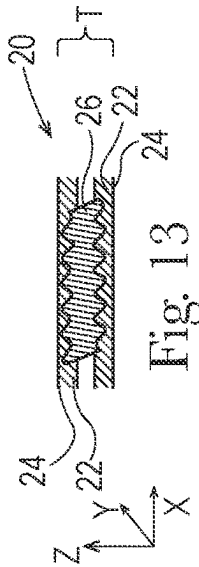


Fig. 13

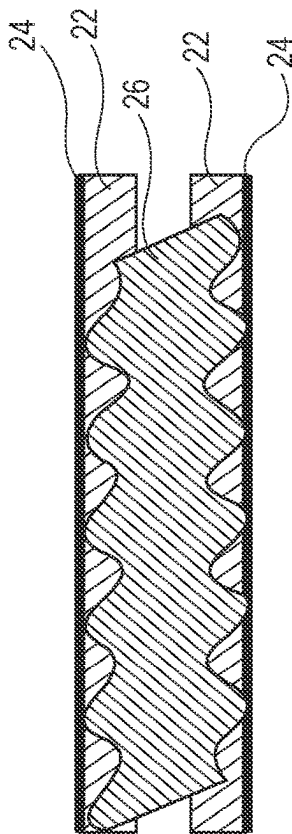


Fig. 14A

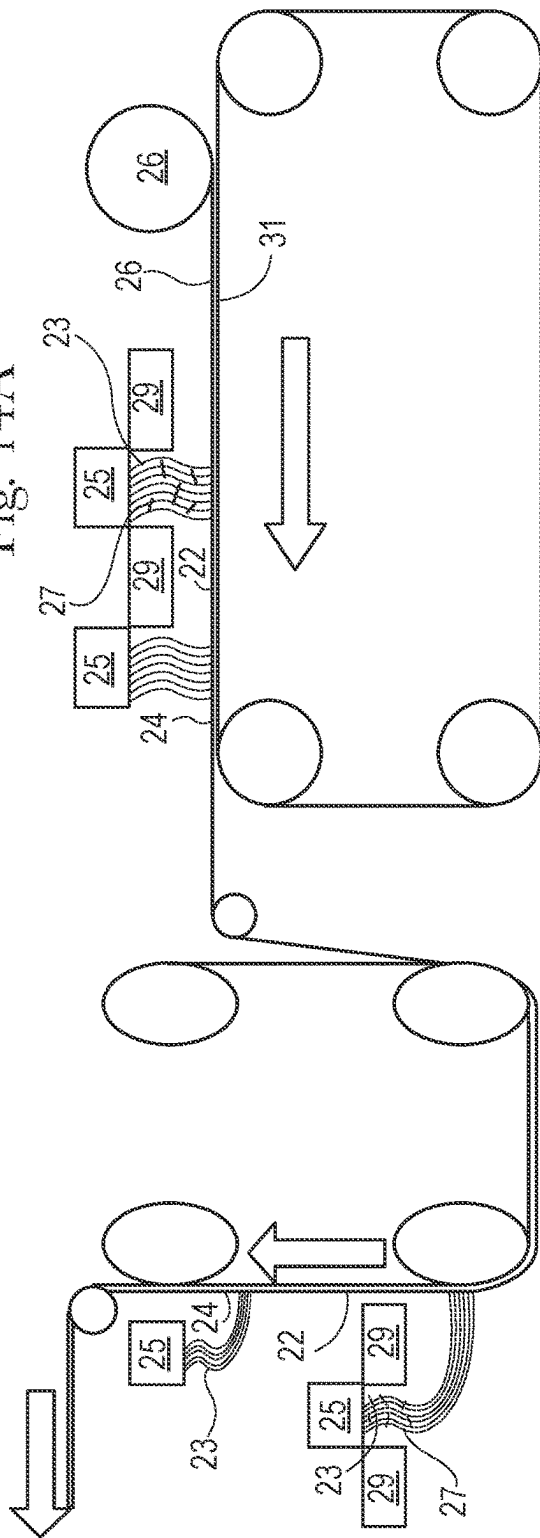


Fig. 14B

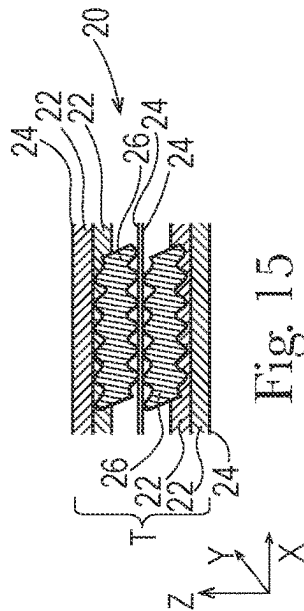
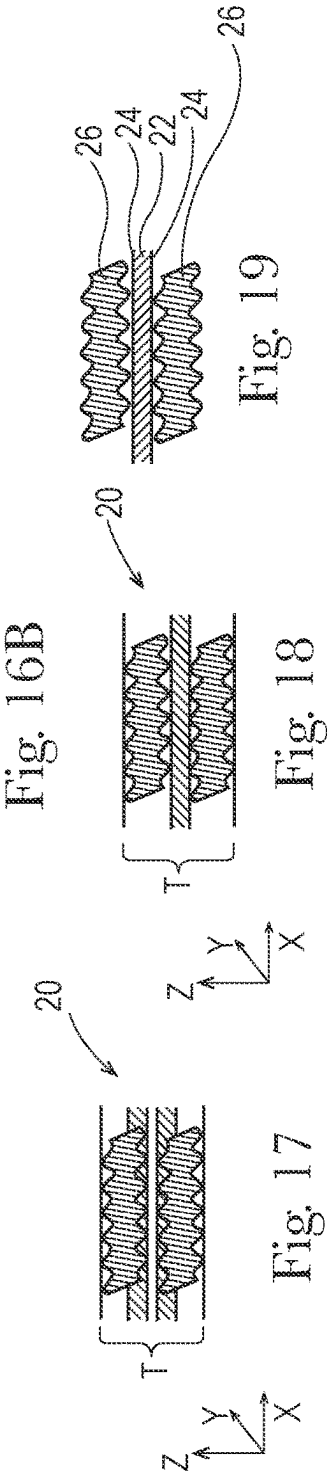
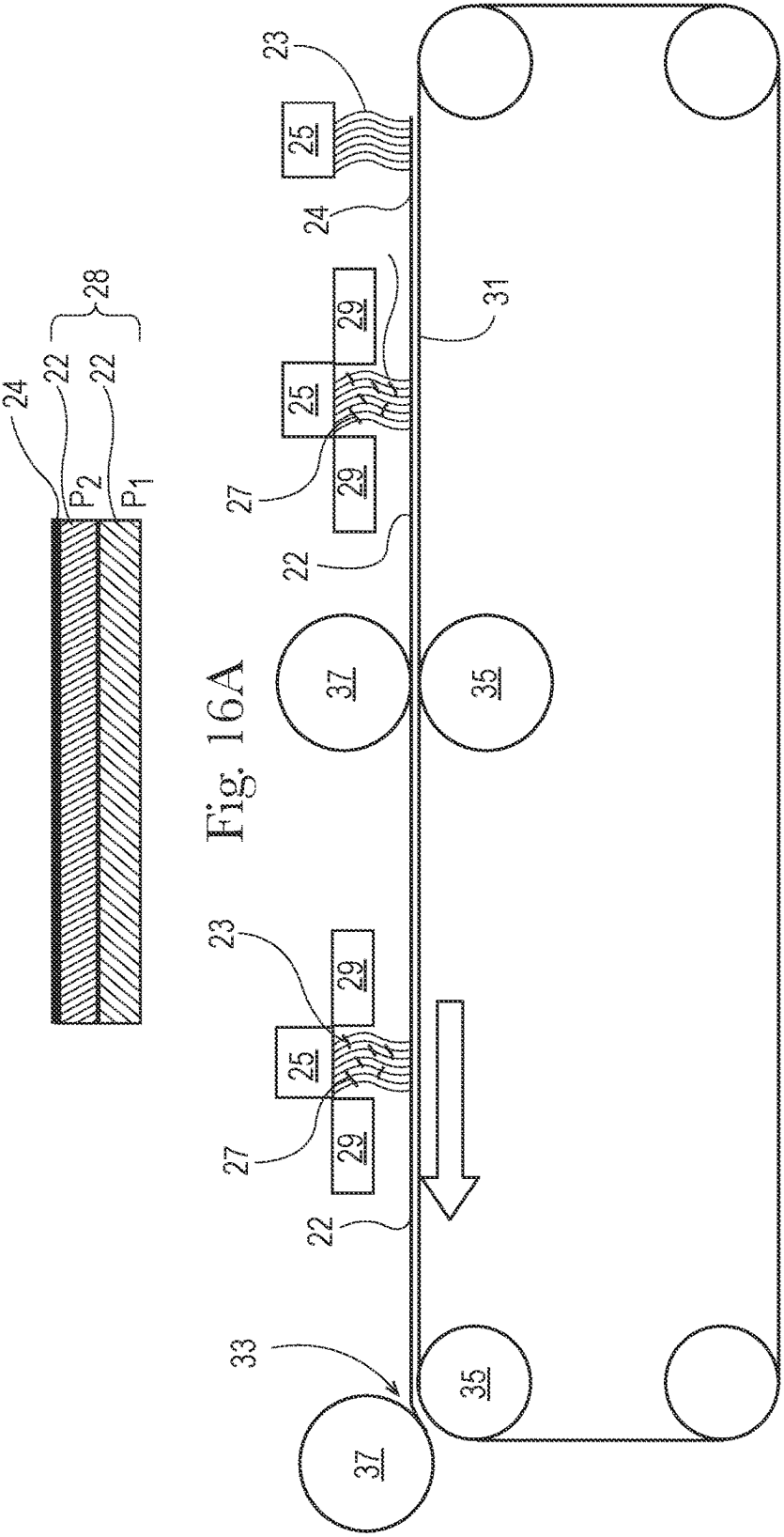


Fig. 15



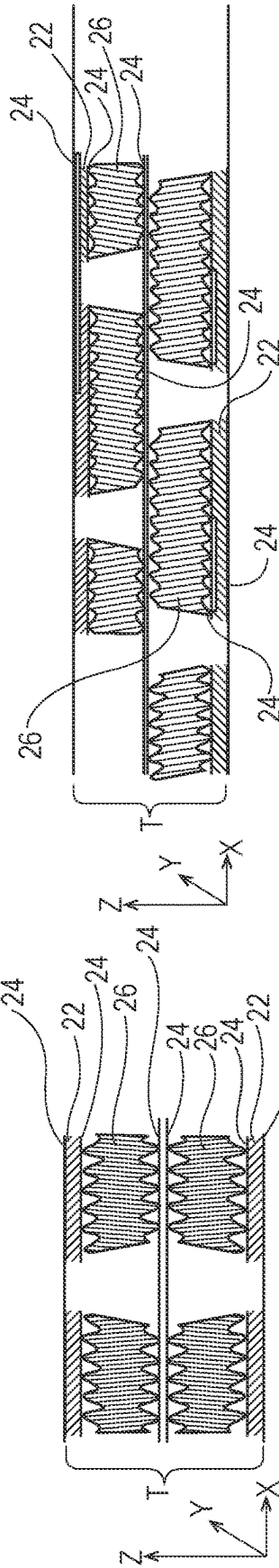


Fig. 20B

Fig. 20A

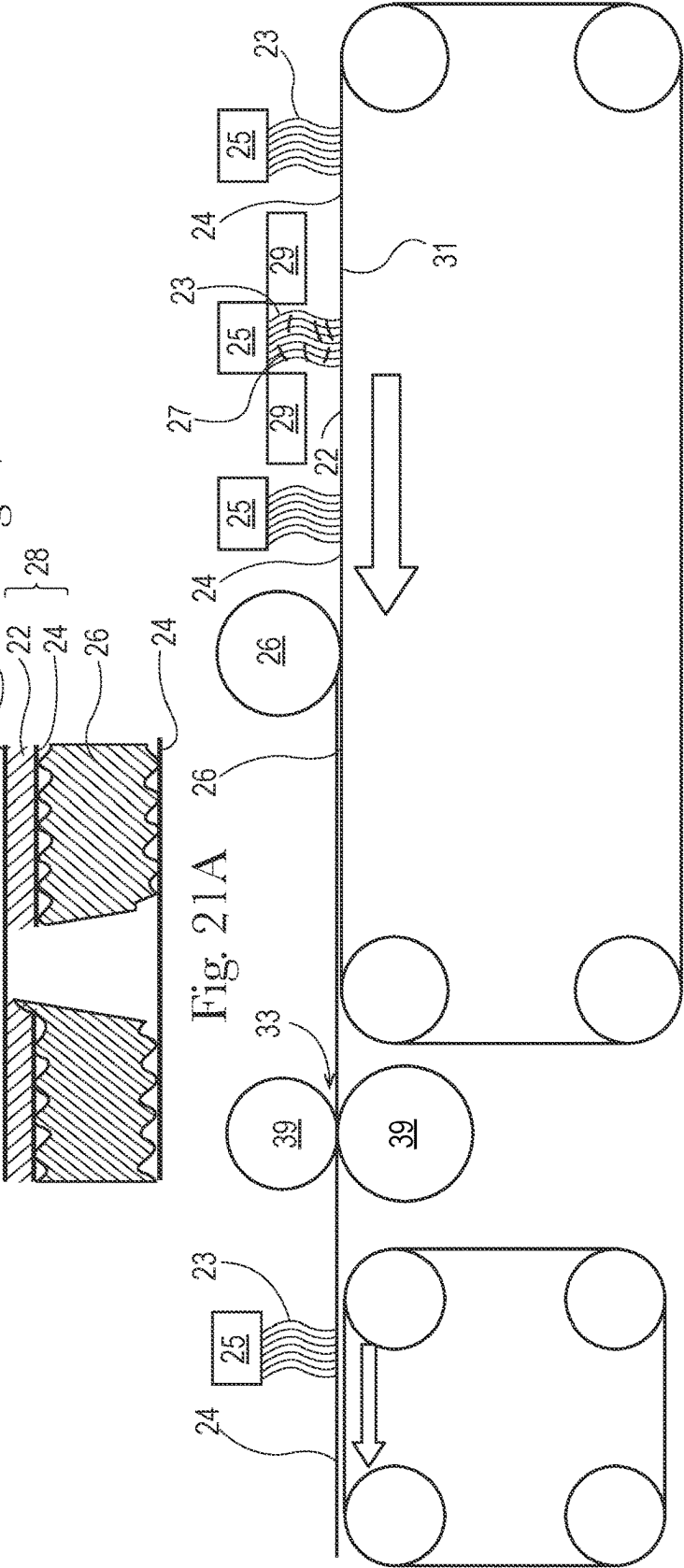


Fig. 21A

Fig. 21B

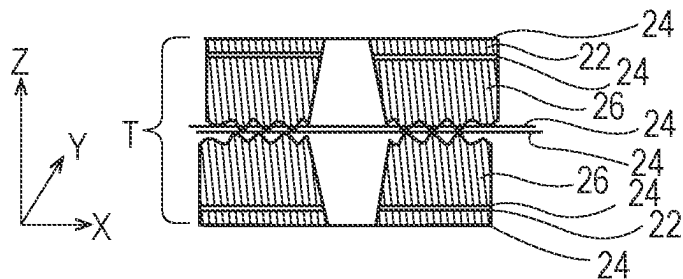


Fig. 22A

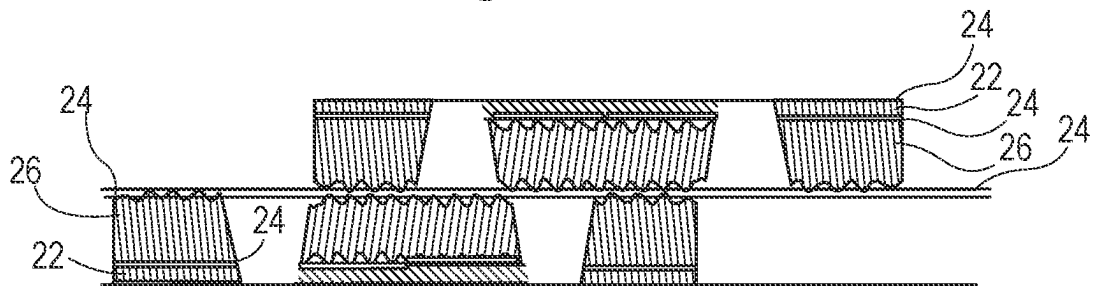


Fig. 22B

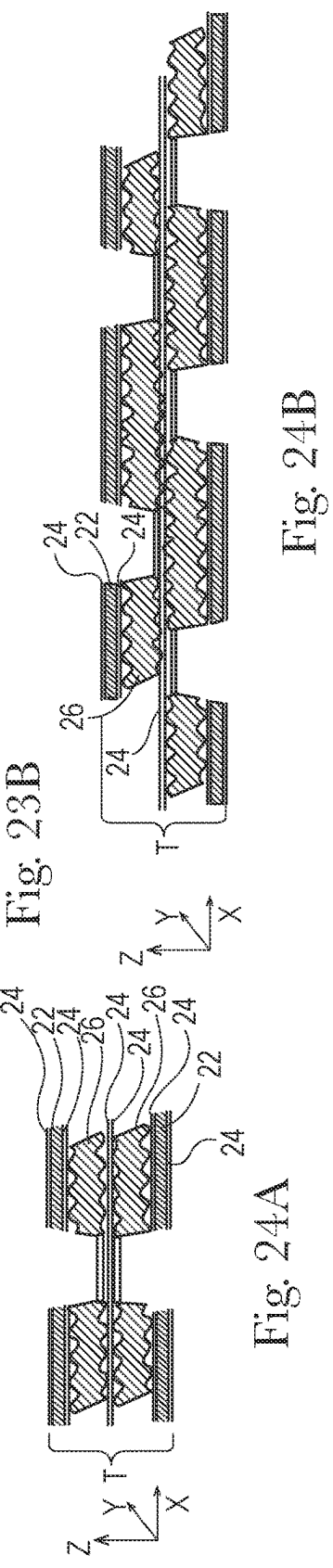
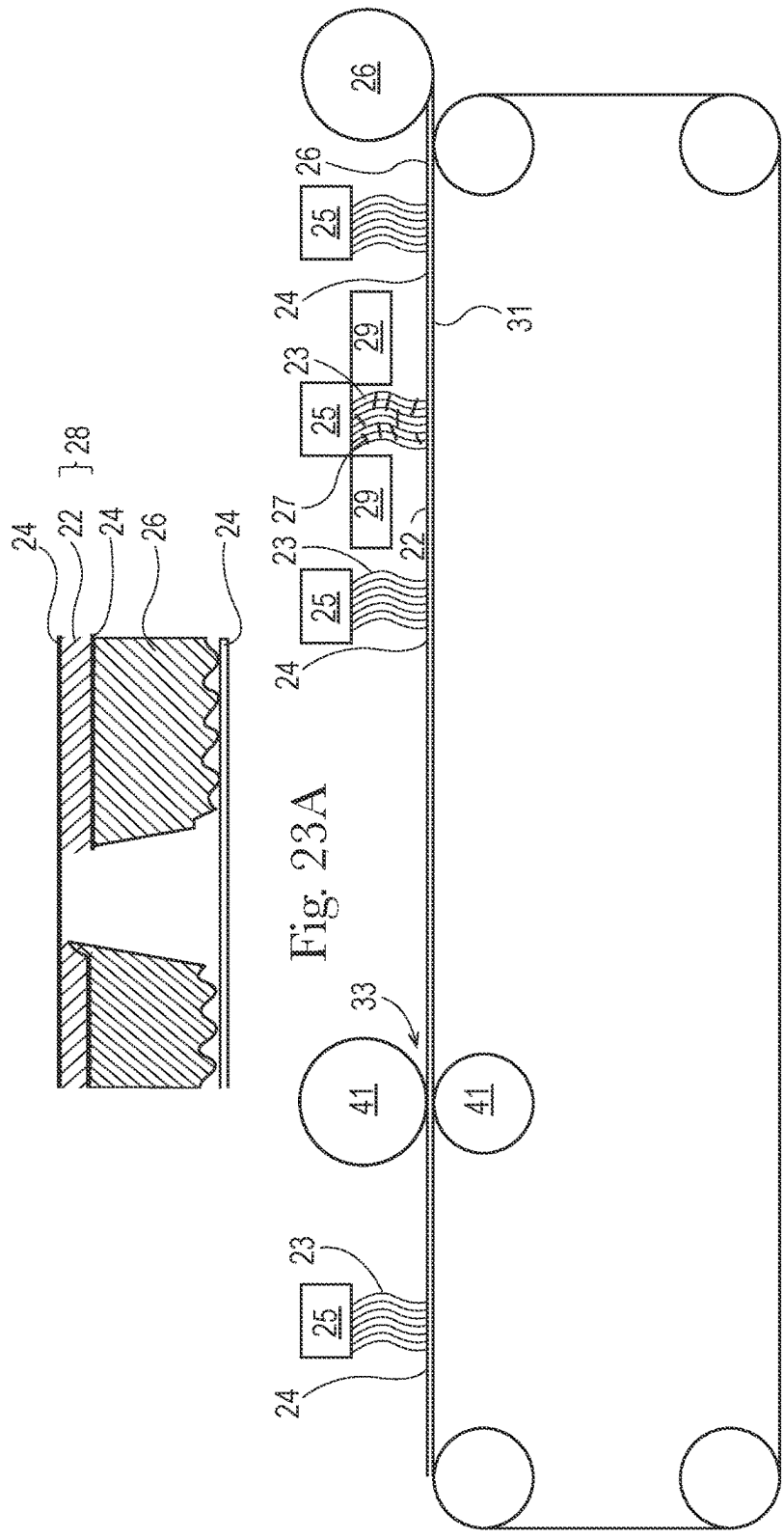


Fig. 23B

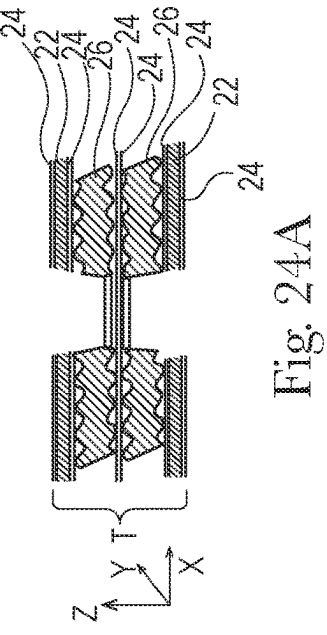


Fig. 24A

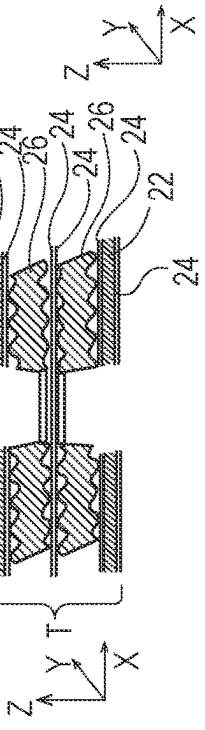


Fig. 24B

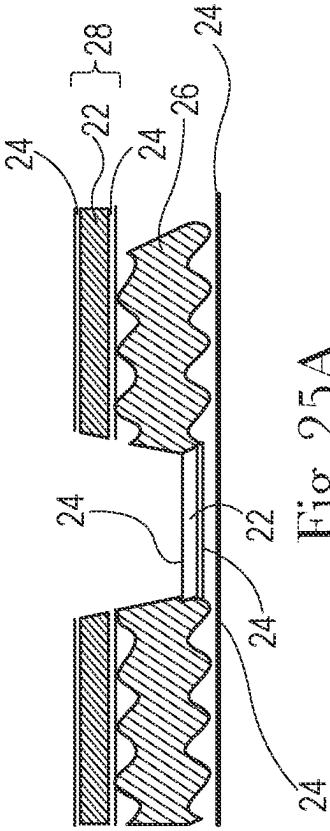


Fig. 25A

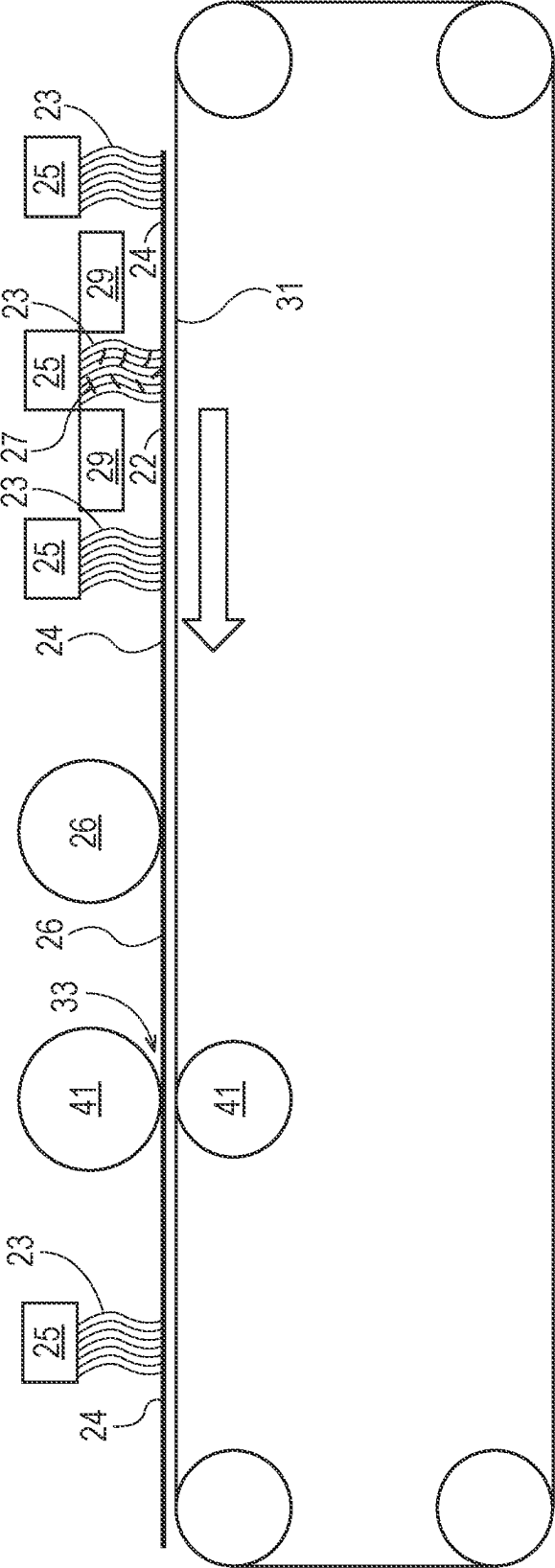


Fig. 25B

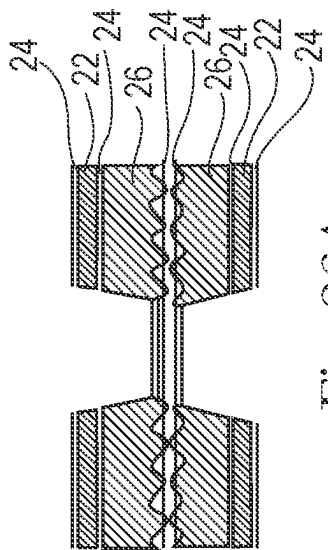


Fig. 26A

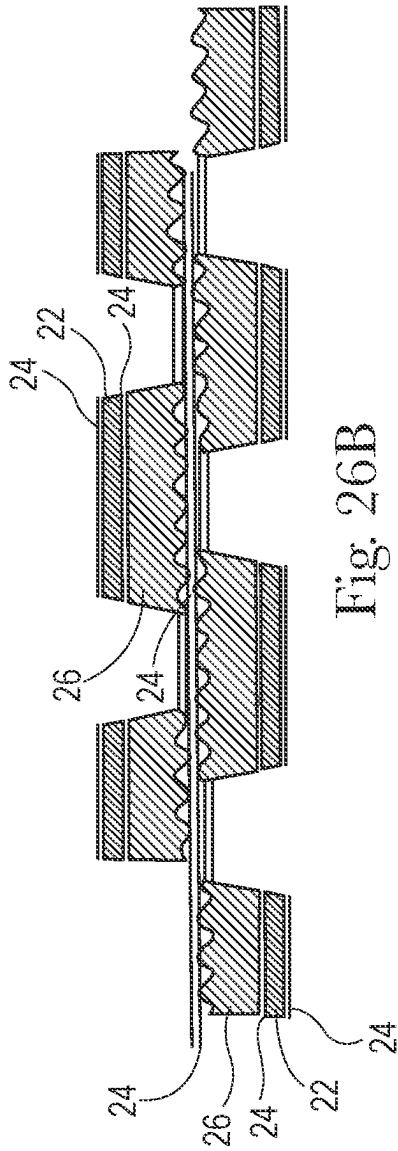


Fig. 26B

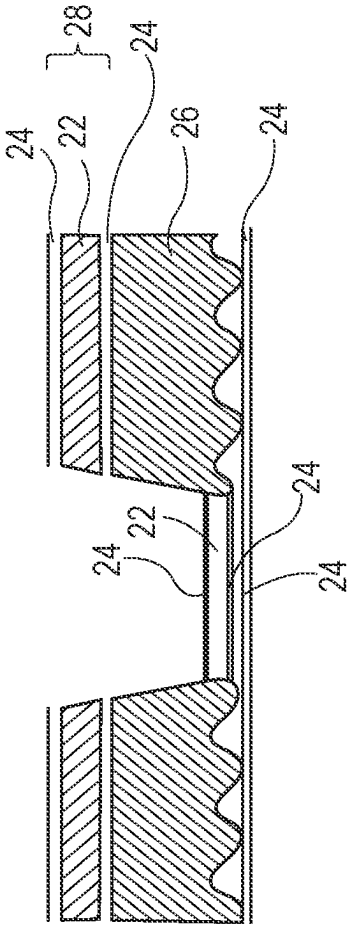


Fig. 27A

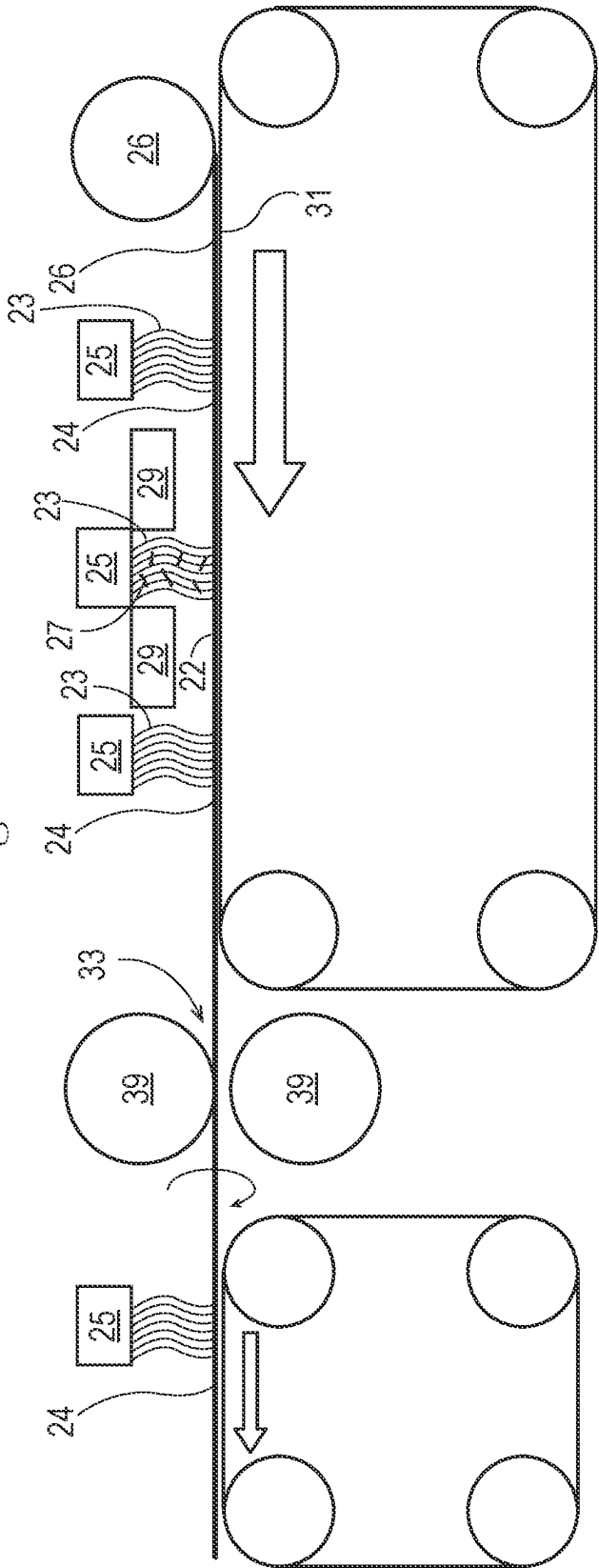


Fig. 27B

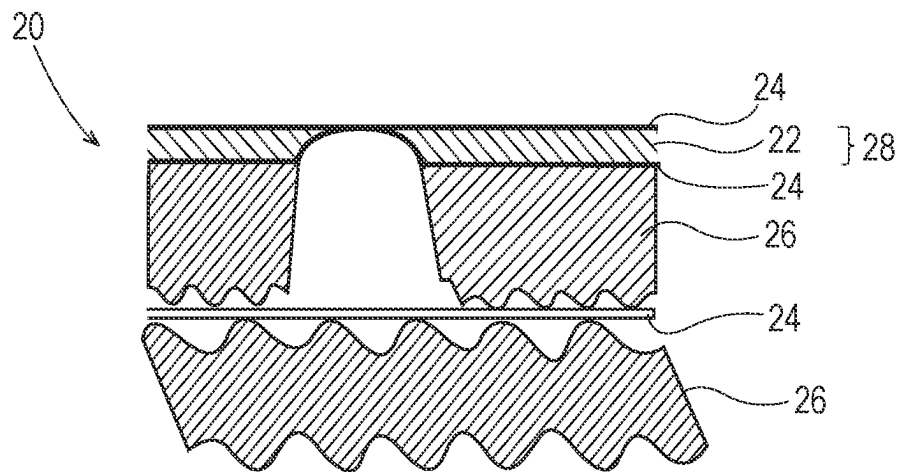


Fig. 28

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FIBROUS STRUCTURE-CONTAINING ARTICLES

FIELD OF THE INVENTION

The present invention relates to articles, such as sanitary tissue products, comprising fibrous structures, and more particularly to articles comprising fibrous structures comprising a plurality of fibrous elements wherein the articles exhibit improved bulk and absorbent properties compared to known articles and methods for making same.

BACKGROUND OF THE INVENTION

Consumers of articles, such as sanitary tissue products, for example paper towels, desire improved roll bulk and/or wet and/or dry sheet bulk compared to known sanitary tissue products, especially paper towels, without negatively impacting the softness and/or stiffness and/or flexibility of the sanitary tissue product. In the past, in order to achieve greater roll bulk and/or wet and/or dry sheet bulk in sanitary issue products, such as paper towels, the softness and/or stiffness and/or flexibility of the sanitary tissue products was negatively impacted.

Consumers of articles, such as sanitary tissue products, for example paper towels, desire improved absorbency compared to known sanitary tissue products, especially paper towels, without negatively impacting the softness and/or stiffness and/or flexibility of the sanitary tissue product. In the past, in order to achieve greater absorbency in sanitary issue products, such as paper towels, the softness and/or stiffness and/or flexibility of the sanitary tissue products were negatively impacted.

Consumers of articles, such as sanitary tissue products, for example paper towels, desire improved absorbency compared to known sanitary tissue products, especially paper towels, without negatively impacting the strength of the sanitary tissue product. In the past, in order to achieve greater absorbency in sanitary issue products, such as paper towels, the strength of the sanitary tissue products was negatively impacted.

Consumers of articles, such as sanitary tissue products, for example paper towels, desire improved hand protection during use compared to known sanitary tissue products, especially paper towels, without negatively impacting absorbency. In the past, in order to achieve greater hand protection in sanitary issue products, such as paper towels, the absorbency of the sanitary tissue products was negatively impacted.

Consumers of articles, such as sanitary tissue products, for example paper towels, desire improved roll bulk and/or wet and/or dry sheet bulk compared to known sanitary tissue products, especially paper towels, without negatively impacting the opacity of the sanitary tissue product. In the past, in order to achieve greater roll bulk and/or wet and/or dry sheet bulk in sanitary issue products, such as paper towels, the opacity of the sanitary tissue products was negatively impacted.

Consumers of articles, such as sanitary tissue products, for example paper towels, desire improved reopenability during use compared to known sanitary tissue products, especially paper towels, without negatively impacting absorbency. In the past, in order to achieve improved reopenability in sanitary issue products, such as paper towels, the absorbency of the sanitary tissue products was negatively impacted.

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Consumers of articles, such as sanitary tissue products, for example paper towels, desire improved absorbency, especially absorbent capacity, compared to known sanitary tissue products, especially paper towels, without negatively impacting the surface drying of the sanitary tissue product. In the past, in order to achieve greater absorbency in sanitary issue products, such as paper towels, the surface drying of the sanitary tissue products was negatively impacted.

Consumers of articles, such as sanitary tissue products, for example paper towels, desire improved wet sheet bulk during use, compared to known sanitary tissue products, especially paper towels, without negatively impacting the surface drying of the sanitary tissue product. In the past, in order to achieve greater wet sheet bulk in sanitary issue products, such as paper towels, the surface drying of the sanitary tissue products was negatively impacted.

In the past, fibers, such as cellulose pulp fibers, have been used in known fibrous structures to achieve bulk and absorbency properties in articles, such as sanitary tissue products, for example paper towels, but such bulk and absorbency properties have been plagued with negatives as described above, such as softness and/or flexibility and/or stiffness negatives and/or the ability to maintain the bulk properties when wet. Examples of such known articles comprising such fibrous structures are described below.

Articles comprising fibrous structures comprising a plurality of fibrous elements, for example filaments and fibers, wherein the articles exhibit differential cellulose content throughout the thickness of the article are known. One prior art article **10** comprising a fibrous structure comprising a plurality of fibrous elements (filaments and/or fibers) as shown in Prior Art FIG. **1** comprises a meltblown or spunbond polymeric abrasive layer **12** and an absorbent layer **14**, such as a wet-laid fibrous structure, a coform fibrous structure, or an air-laid fibrous structure. In one example, the cellulose content throughout the thickness *T* (along the *z*-axis) of the prior art article **10** when the absorbent layer **14** is a wet-laid or air-laid fibrous structure is such that a first portion, for example the abrasive layer **12**, of the prior art article **10** exhibits a cellulose content of less than 40%, for example about 0% by weight of the fibrous elements in the first portion, and a second portion of the prior art article **10**, for example the absorbent layer **14**; namely, the wet-laid or air-laid fibrous structure, exhibits a cellulose content of 95% to 100%, for example 100% by weight of the fibrous elements in the second portion.

In another example of Prior Art FIG. **1**, the cellulose content throughout the thickness *T* of the prior art article **10** when the absorbent layer **14** is a coform fibrous structure is such that a first portion, for example the abrasive layer **12**, of the prior art article **10** exhibits a cellulose content of less than 40%, for example about 0% by weight of the fibrous elements in the first portion, and a second portion, for example the absorbent layer **14**; namely, the coform fibrous structure, exhibits a cellulose content of 40% to less than 95% by weight of the fibrous elements in the second portion.

As shown in Prior Art FIG. **1**, the prior art article **10** fails to teach a cellulose content such that the cellulose content of a first portion of the prior art article **10** is from 0% to less than 40% by weight of the fibrous elements in the first portion, the cellulose content of a second portion of the prior art article **10** different from the first portion is from 40% to less than 93% by weight of the fibrous elements in the second portion, and the cellulose content of a third portion of the prior art article **10** different from the first and second portions is 93% to 100% by weight of the fibrous elements

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in the third portion, and wherein at least the second portion comprises a mixture of filaments and fibers.

Accordingly, there is a need for articles comprising fibrous structures that exhibit novel differential cellulose content that results in the articles exhibiting improved bulk and/or absorbent properties that are consumer acceptable that maintain sufficient such bulk properties when wet during use by consumers and/or without negatively impacting and/or improving the softness and/or flexibility and/or stiffness of such articles and methods for making same.

SUMMARY OF THE INVENTION

The present invention fulfills the need described above by providing articles comprising fibrous structures that exhibit novel cellulose contents such that the articles exhibit improved bulk and/or absorbent properties that are consumer acceptable while still maintaining such bulk properties when wet and/or without negatively impacting and/or improving the softness and/or flexibility and/or stiffness of such articles and methods for making same.

One solution to the problem identified above are articles, such as sanitary tissue products, for example paper towels, that comprise fibrous structures that utilize a plurality of fibrous elements, such as filaments and/or fibers, arranged within the articles such that the articles exhibit cellulose contents, such as within the fibrous elements, for example as cellulose pulp fibers (e.g., wood pulp fibers), that vary throughout the thickness of the articles containing such fibrous structure such that the cellulose content of a first portion of an article is from 0% to less than 40% by weight of the fibrous elements in the first portion (which by default herein means the remainder of fibrous elements present within the first portion do not contain cellulose, for example contain a synthetic polymer, such as a thermoplastic polymer like polypropylene), the cellulose content of a second portion of the article different from the first portion is from 40% to less than 95% by weight of the fibrous elements in the second portion, and the cellulose content of a third portion of the article different from the first and second portions is 95% to 100% by weight of the fibrous elements in the third portion, and wherein at least the second portion comprises a mixture of filaments and fibers. Such an arrangement of cellulose content within the article as described above results in the article exhibiting improved bulk and/or absorbency compared to known fibrous structures while still maintaining or at least maintaining more of the bulk properties when wet compared to known properties and/or without negatively impacting and/or improving the softness and/or flexibility and/or stiffness properties of the article compared to known articles comprising fibrous structures.

It has unexpectedly been found that the arrangement of the fibrous structures and/or fibrous webs (fibrous web plies) within the articles of the present invention and/or type of fibrous structures and/or type of fibrous elements, for example filaments and/or fibers, within the articles of the present invention result in the article of the present invention exhibiting novel properties, such as bulk and/or absorbent properties without negatively impacting the softness and/or flexibility and/or stiffness of the articles.

In one example of the present invention, an article comprising:

- a. a paper web; and
- b. a multi-component fibrous structure web comprising a plurality of filaments and a plurality of solid additives;

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wherein the multi-component fibrous web is associated with the paper web, is provided.

In another example of the present invention, an article comprising:

- a. a mono-fibrous element web, for example a paper web, comprising a plurality of fibers; and
 - b. a multi-fibrous element web comprising a plurality of filaments and a plurality of fibers;
- wherein the multi-fibrous element web is associated with the mono-fibrous element web, is provided.

In another example of the present invention, an article comprising:

- a. a wet-laid fibrous structure and/or wet-laid fibrous web; and
 - b. a multi-fibrous element web comprising a plurality of filaments and a plurality of fibers;
- wherein the multi-fibrous element web is associated with the wet-laid fibrous structure and/or wet-laid fibrous web, is provided.

In still another example of the present invention, an article comprising:

- a. a mono-fibrous element web, for example a paper web; and
 - b. a multi-fibrous element web, for example a co-formed fibrous structure and/or co-formed fibrous web;
- wherein the multi-fibrous element web is associated with the mono-fibrous element web, is provided.

In another example of the present invention, an article comprising:

- a. a mono-fibrous element wet-laid fibrous structure and/or wet-laid fibrous web; and
 - b. a multi-fibrous element web, for example co-formed fibrous structure and/or co-formed fibrous web;
- wherein the multi-fibrous element web is associated with the mono-fibrous element wet-laid fibrous web along an interface comprising the mono-fibrous element wet-laid fibrous web and the multi-fibrous element web, is provided.

In another example of the present invention, an article comprising:

- a. a mono-fibrous element fibrous web, for example a paper web; and
 - b. a multi-fibrous element fibrous web, for example a co-formed fibrous structure and/or co-formed fibrous web;
- wherein the multi-fibrous element fibrous web is associated with the mono-fibrous element fibrous web along an interface comprising the mono-fibrous element fibrous web and the multi-fibrous element fibrous web, is provided.

In another example of the present invention, an article comprising:

- a. a wet-laid fibrous structure and/or wet-laid fibrous web;
- b. a co-formed fibrous structure and/or co-formed fibrous web; and
- c. a fibrous structure different from the wet-laid fibrous web and the co-formed fibrous structure and/or co-formed fibrous web, is provided.

The present invention provides novel articles comprising fibrous structures comprising fibrous elements that result in the articles exhibiting novel bulk and/or absorbent properties and methods for making same.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a cross-sectional representation of an example of a prior art article.

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FIG. 2A is a cross-sectional representation of an example of a co-formed fibrous structure according to the present invention;

FIG. 2B is an example of a process for making the co-formed fibrous structure of FIG. 2A;

FIG. 3 is a cross-sectional representation of an example of an article according to the present invention;

FIG. 4 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 5 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 6A is a cross-sectional representation of another example of a fibrous web according to the present invention;

FIG. 6B is an example of a process for making the fibrous web of FIG. 6A;

FIG. 7 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 8 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 9A is a cross-sectional representation of another example of an article according to the present invention;

FIG. 9B is an example of a process for making the article according to FIG. 9A

FIG. 10 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 11 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 12 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 13 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 14A is a cross-sectional representation of another example of an article according to the present invention;

FIG. 14B is an example of a process for making the article of FIG. 14A;

FIG. 15 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 16A is a cross-sectional representation of another example of an article according to the present invention;

FIG. 16B is an example of a process for making the article of FIG. 16A;

FIG. 17 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 18 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 19 is a cross-sectional representation of another example of an article according to the present invention;

FIG. 20A is a cross-sectional representation of another example of an article according to the present invention;

FIG. 20B is a cross-sectional representation of another example of an article according to the present invention;

FIG. 21A is a cross-sectional representation of another example of a fibrous web according to the present invention suitable for use in the article of FIGS. 20A and 20B;

FIG. 21B is an example of a process for making the fibrous web of FIG. 21A;

FIG. 22A is a cross-sectional representation of another example of an article according to the present invention;

FIG. 22B is a cross-sectional representation of another example of an article according to the present invention;

FIG. 23A is a cross-sectional representation of another example of a fibrous web according to the present invention suitable for use in the article of FIGS. 22A and 22B;

FIG. 23B is an example of a process for making the fibrous web of FIG. 23A;

FIG. 24A is a cross-sectional representation of another example of an article according to the present invention;

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FIG. 24B is a cross-sectional representation of another example of an article according to the present invention;

FIG. 25A is a cross-sectional representation of another example of a fibrous web according to the present invention suitable for use in the article of FIGS. 24A and 24B;

FIG. 25B is an example of a process for making the fibrous web of FIG. 25A;

FIG. 26A is a cross-sectional representation of another example of an article according to the present invention;

FIG. 26B is a cross-sectional representation of another example of an article according to the present invention;

FIG. 27A is a cross-sectional representation of another example of a fibrous web according to the present invention suitable for use in the article of FIGS. 26A and 26B;

FIG. 27B is an example of a process for making the fibrous web of FIG. 27A; and

FIG. 28 is a cross-section representation of another example of an article according to the present invention.

DETAILED DESCRIPTION OF THE INVENTION

“Article” as used herein means a consumer-usable structure comprising one or more and/or two or more and/or three or more and/or four or more fibrous webs according to the present invention. In one example the article is a dry article. In addition, the article may be a sanitary tissue product. The article may comprise two or more and/or three or more different fibrous webs selected from the group consisting of: wet-laid fibrous webs, air-laid fibrous webs, co-formed fibrous web, meltblown fibrous web, and spunbond fibrous web. In one example, the article is void of a hydroentangled fibrous web and/or is not a hydroentangled fibrous web. In another example, the article is void of a carded fibrous web and/or is not a carded fibrous web. In addition to the fibrous webs, the articles of the present invention may comprise other solid matter, such as sponges, foams, particle, such as absorbent gel materials, and mixtures thereof.

In one example, two or more fibrous webs (fibrous web plies) of the present invention may be associated together to form the article.

In one example, the article of the present invention comprises one or more co-formed fibrous webs (co-formed fibrous web plies). In addition to the co-formed fibrous web, the article may further comprise one or more wet-laid fibrous webs (wet-laid fibrous web plies). Also in addition to the co-formed fibrous web (co-formed fibrous web ply) with or without one or more wet-laid fibrous webs (wet-laid fibrous web plies), the article may further comprise one or more meltblown fibrous webs (meltblown fibrous web plies).

In another example, the article of the present invention may comprise one or more multi-fibrous element fibrous webs (e.g., a fibrous structure comprising a mixture of filaments and fibers), such as a co-formed fibrous web, and one or more mono-fibrous element fibrous webs (e.g., a fibrous structure comprising only fibers or only filaments, not a mixture of fibers and filaments), such as a paper web, for example a fibrous web and/or a meltblown fibrous web.

In one example, at least a portion of the article exhibits a basis weight of about 150 gsm or less and/or about 100 gsm or less and/or from about 30 gsm to about 95 gsm.

“Sanitary tissue product” as used herein means a soft, low density (i.e. <about 0.15 g/cm³) web useful as a wiping implement for post-urinary and post-bowel movement cleaning (toilet tissue), for otorhinolaryngological discharges (facial tissue), and multi-functional absorbent and

cleaning uses (absorbent towels). Non-limiting examples of suitable sanitary tissue products of the present invention include paper towels, bath tissue, facial tissue, napkins, baby wipes, adult wipes, wet wipes, cleaning wipes, polishing wipes, cosmetic wipes, car care wipes, wipes that comprise an active agent for performing a particular function, cleaning substrates for use with implements, such as a Swiffer® cleaning wipe/pad. The sanitary tissue product may be convolutedly wound upon itself about a core or without a core to form a sanitary tissue product roll.

The sanitary tissue products of the present invention may exhibit a basis weight between about 10 g/m² to about 500 g/m² and/or from about 15 g/m² to about 400 g/m² and/or from about 20 g/m² to about 300 g/m² and/or from about 20 g/m² to about 200 g/m² and/or from about 20 g/m² to about 150 g/m² and/or from about 20 g/m² to about 120 g/m² and/or from about 20 g/m² to about 110 g/m² and/or from about 20 g/m² to about 100 g/m² and/or from about 30 to 90 g/m². In addition, the sanitary tissue product of the present invention may exhibit a basis weight between about 40 g/m² to about 500 g/m² and/or from about 50 g/m² to about 400 g/m² and/or from about 55 g/m² to about 300 g/m² and/or from about 60 to 200 g/m². In one example, the sanitary tissue product exhibits a basis weight of less than 100 g/m² and/or less than 80 g/m² and/or less than 75 g/m² and/or less than 70 g/m² and/or less than 65 g/m² and/or less than 60 g/m² and/or less than 55 g/m² and/or less than 50 g/m² and/or less than 47 g/m² and/or less than 45 g/m² and/or less than 40 g/m² and/or less than 35 g/m² and/or to greater than 20 g/m² and/or greater than 25 g/m² and/or greater than 30 g/m² as measured according to the Basis Weight Test Method described herein.

The sanitary tissue products of the present invention may exhibit a density (measured at 95 g/in²) of less than about 0.60 g/cm³ and/or less than about 0.30 g/cm³ and/or less than about 0.20 g/cm³ and/or less than about 0.10 g/cm³ and/or less than about 0.07 g/cm³ and/or less than about 0.05 g/cm³ and/or from about 0.01 g/cm³ to about 0.20 g/cm³ and/or from about 0.02 g/cm³ to about 0.10 g/cm³.

The sanitary tissue products of the present invention may comprises additives such as softening agents, temporary wet strength agents, permanent wet strength agents, bulk softening agents, silicones, wetting agents, latexes, especially surface-pattern-applied latexes, dry strength agents such as carboxymethylcellulose and starch, and other types of additives suitable for inclusion in and/or on sanitary tissue products.

"Fibrous web" as used herein means a unitary structure comprising one or more fibrous structures that are associated with one another, such as by compression bonding (for example by passing through a nip formed by two rollers), thermal bonding (for example by passing through a nip formed by two rollers where at least one of the rollers is heated to a temperature of at least about 120° C. (250° F.), microselfing, needle punching, and gear rolling, to form the unitary structure, for example a unitary structure that exhibits sufficient integrity to be processed with web handling equipment and/or exhibits a basis weight of at least 6 gsm and/or at least 8 gsm and/or at least 10 gsm and/or at least 15 gsm and/or at least 20 gsm and/or at least 30 gsm and/or at least 40 gsm. The unitary structure may also be referred to as a ply, a fibrous web ply.

"Fibrous structure" as used herein means a structure that comprises a plurality of fibrous elements, for example a plurality of filaments and/or a plurality of fibers, for example pulp fibers, for example wood pulp fibers, and/or cellulose fibrous elements and/or cellulose fibers, such as pulp fibers,

for example wood pulp fibers. In addition to the fibrous elements, the fibrous structures may comprise particles, such as absorbent gel material particles. In one example, a fibrous structure according to the present invention means an orderly arrangement of fibrous elements within a structure in order to perform a function. In another example, a fibrous structure according to the present invention is a nonwoven. In one example, the fibrous structures of the present invention may comprise wet-laid fibrous structures, for example embossed conventional wet pressed fibrous structures, through-air-dried (TAD) fibrous structures both creped and/or uncreped, belt-creped fibrous structures, fabric-creped fibrous structures, and combinations thereof, air-laid fibrous structures, such as thermally-bonded air-laid (TBAL) fibrous structures, melt-bonded air-laid (MBAL), latex-bonded air-laid (LBAL) fibrous structures and combinations thereof, co-formed fibrous structures, meltblown fibrous structures, and spunbond fibrous structures, carded fibrous structures, and combinations thereof. In one example, the fibrous structure is a non-hydroentangled fibrous structure. In another example, the fibrous structure is a non-carded fibrous structure.

In another example of the present invention, a fibrous structure comprises a plurality of inter-entangled fibrous elements, for example inter-entangled filaments.

Non-limiting examples of fibrous structures and/or fibrous webs (fibrous web plies) of the present invention include paper.

The fibrous structures of the present invention may be homogeneous or may be layered. If layered, the fibrous structures may comprise at least two and/or at least three and/or at least four and/or at least five layers.

Any one of the fibrous structures may itself be a fibrous web (fibrous web ply) if the fibrous structure exhibits sufficient integrity to be processed with web handling equipment and/or exhibits a basis weight of at least 6 gsm and/or at least 8 gsm and/or at least 10 gsm and/or at least 15 gsm and/or at least 20 gsm and/or at least 30 gsm and/or at least 40 gsm. An example of such a fibrous structure, for example a paper web, for example a fibrous structure exhibiting a basis weight of at least 10 gsm and/or at least 15 gsm and/or at least 20 gsm can be a fibrous web (fibrous web ply) itself.

Non-limiting examples of processes for making the fibrous structures of the present invention include known wet-laid papermaking processes, for example conventional wet-pressed (CWP) papermaking processes and through-air-dried (TAD), both creped TAD and uncreped TAD, papermaking processes, and air-laid papermaking processes. Such processes typically include steps of preparing a fiber composition in the form of a fiber suspension in a medium, either wet, more specifically aqueous medium, or dry, more specifically gaseous, i.e. with air as medium. The aqueous medium used for wet-laid processes is oftentimes referred to as a fiber slurry. The fiber slurry is then used to deposit a plurality of the fibers onto a forming wire, fabric, or belt such that an embryonic web material is formed, after which drying and/or bonding the fibers together results in a fibrous structure and/or fibrous web (fibrous web ply). Further processing of the fibrous structure and/or fibrous web (fibrous web ply) may be carried out such that a fibrous structure and/or fibrous web (fibrous web ply) is formed. For example, in typical papermaking processes, the fibrous structure and/or fibrous web (fibrous web ply) is wound on the reel at the end of papermaking, often referred to as a parent roll, and may subsequently be converted into a fibrous web (fibrous web ply) of the present invention and/or

ultimately incorporated into an article, such as a single- or multi-ply sanitary tissue product.

“Multi-fibrous element fibrous structure” as used herein means a fibrous structure that comprises filaments and fibers, for example a co-formed fibrous structure is a multi-fibrous element fibrous structure.

“Mono-fibrous element fibrous structure” as used herein means a fibrous structure that comprises only fibers or filaments, for example a paper web, such as a paper web, for example a fibrous structure, or meltblown fibrous structure, such as a scrim, respectively, not a mixture of fibers and filaments.

“Co-formed fibrous structure” as used herein means that the fibrous structure comprises a mixture of filaments, for example meltblown filaments, such as thermoplastic filaments, for example polypropylene filaments, and fibers, such as pulp fibers, for example wood pulp fibers. The filaments and fibers are commingled together to form the co-formed fibrous structure. The co-formed fibrous structure may be associated with one or more meltblown fibrous structures and/or spunbond fibrous structures, which form a scrim (in one example the scrim may be present at a basis weight of greater than 0.5 gsm to about 5 gsm and/or from about 1 gsm to about 4 gsm and/or from about 1 gsm to about 3 gsm and/or from about 1.5 gsm to about 2.5 gsm), such as on one or more surfaces of the co-formed fibrous structure.

The co-formed fibrous structure of the present invention may be made via a co-forming process. A non-limiting example of making a co-formed fibrous structure and/or co-formed fibrous web (co-formed fibrous web ply) comprising a co-formed fibrous structure associated with or without a meltblown fibrous structure, for example a scrim layer of filaments, on one or both surfaces, when present, of the co-formed fibrous structure and process for making is shown in FIGS. 2A and 2B.

“Fibrous element” as used herein means an elongate particulate having a length greatly exceeding its average diameter, i.e. a length to average diameter ratio of at least about 10. A fibrous element may be a filament or a fiber. In one example, the fibrous element is a single fibrous element rather than a yarn comprising a plurality of fibrous elements.

The fibrous elements of the present invention may be spun from polymer melt compositions via suitable spinning operations, such as meltblowing and/or spunbonding and/or they may be obtained from natural sources such as vegetative sources, for example trees.

The fibrous elements of the present invention may be monocomponent and/or multicomponent. For example, the fibrous elements may comprise bicomponent fibers and/or filaments. The bicomponent fibers and/or filaments may be in any form, such as side-by-side, core and sheath, islands-in-the-sea and the like.

“Filament” as used herein means an elongate particulate as described above that exhibits a length of greater than or equal to 5.08 cm (2 in.) and/or greater than or equal to 7.62 cm (3 in.) and/or greater than or equal to 10.16 cm (4 in.) and/or greater than or equal to 15.24 cm (6 in.).

Filaments are typically considered continuous or substantially continuous in nature. Filaments are relatively longer than fibers. Non-limiting examples of filaments include meltblown and/or spunbond filaments. Non-limiting examples of polymers that can be spun into filaments include natural polymers, such as starch, starch derivatives, cellulose, such as rayon and/or lyocell, and cellulose derivatives, hemicellulose, hemicellulose derivatives, and synthetic polymers including, but not limited to polyvinyl

alcohol filaments and/or polyvinyl alcohol derivative filaments, and thermoplastic polymer filaments, such as polyesters, nylons, polyolefins such as polypropylene filaments, polyethylene filaments, and biodegradable or compostable thermoplastic fibers such as polylactic acid filaments, polyhydroxyalkanoate filaments, polyesteramide filaments, and polycaprolactone filaments. The filaments may be monocomponent or multicomponent, such as bicomponent filaments.

The filaments may be made via spinning, for example via meltblowing and/or spunbonding, from a polymer, for example a thermoplastic polymer, such as polyolefin, for example polypropylene and/or polyethylene, and/or polyester. Filaments are typically considered continuous or substantially continuous in nature.

“Meltblowing” is a process for producing filaments directly from polymers or resins using high-velocity air or another appropriate force to attenuate the filaments before collecting the filaments on a collection device, such as a belt, for example a patterned belt or molding member. In a meltblowing process the attenuation force is applied in the form of high speed air as the material (polymer) exits a die or spinnerette.

“Spunbonding” is a process for producing filaments directly from polymers by allowing the polymer to exit a die or spinnerette and drop a predetermined distance under the forces of flow and gravity and then applying a force via high velocity air or another appropriate source to draw and/or attenuate the polymer into a filament.

“Fiber” as used herein means an elongate particulate as described above that exhibits a length of less than 5.08 cm (2 in.) and/or less than 3.81 cm (1.5 in.) and/or less than 2.54 cm (1 in.).

Fibers are typically considered discontinuous in nature. Non-limiting examples of fibers include pulp fibers, such as wood pulp fibers, and synthetic staple fibers such as polypropylene, polyethylene, polyester, copolymers thereof, rayon, lyocell, glass fibers and polyvinyl alcohol fibers.

Staple fibers may be produced by spinning a filament tow and then cutting the tow into segments of less than 5.08 cm (2 in.) thus producing fibers; namely, staple fibers.

“Pulp fibers” as used herein means fibers that have been derived from vegetative sources, such as plants and/or trees. In one example of the present invention, “pulp fiber” refers to papermaking fibers. In one example of the present invention, a fiber may be a naturally occurring fiber, which means it is obtained from a naturally occurring source, such as a vegetative source, for example a tree and/or plant, such as trichomes. Such fibers are typically used in papermaking and are oftentimes referred to as papermaking fibers. Papermaking fibers useful in the present invention include cellulosic fibers commonly known as wood pulp fibers. Applicable wood pulps include chemical pulps, such as Kraft, sulfite, and sulfate pulps, as well as mechanical pulps including, for example, groundwood, thermomechanical pulp and chemically modified thermomechanical pulp. Chemical pulps, however, may be preferred since they impart a superior tactile sense of softness to fibrous structures made therefrom. Pulps derived from both deciduous trees (hereinafter, also referred to as “hardwood”) and coniferous trees (hereinafter, also referred to as “softwood”) may be utilized. The hardwood and softwood fibers can be blended, or alternatively, can be deposited in layers to provide a stratified web. Also applicable to the present invention are fibers derived from recycled paper, which may contain any or all of the above categories of fibers as well as

other non-fibrous polymers such as fillers, softening agents, wet and dry strength agents, and adhesives used to facilitate the original papermaking.

In one example, the wood pulp fibers are selected from the group consisting of hardwood pulp fibers, softwood pulp fibers, and mixtures thereof. The hardwood pulp fibers may be selected from the group consisting of: tropical hardwood pulp fibers, northern hardwood pulp fibers, and mixtures thereof. The tropical hardwood pulp fibers may be selected from the group consisting of: *eucalyptus* fibers, acacia fibers, and mixtures thereof. The northern hardwood pulp fibers may be selected from the group consisting of: cedar fibers, maple fibers, and mixtures thereof.

In addition to the various wood pulp fibers, other cellulosic fibers such as cotton linters, rayon, lyocell, trichomes, seed hairs, rice straw, wheat straw, bamboo, and bagasse fibers can be used in this invention. Other sources of cellulose in the form of fibers or capable of being spun into fibers include grasses and grain sources.

“Trichome” or “trichome fiber” as used herein means an epidermal attachment of a varying shape, structure and/or function of a non-seed portion of a plant. In one example, a trichome is an outgrowth of the epidermis of a non-seed portion of a plant. The outgrowth may extend from an epidermal cell. In one embodiment, the outgrowth is a trichome fiber. The outgrowth may be a hairlike or bristle-like outgrowth from the epidermis of a plant.

Trichome fibers are different from seed hair fibers in that they are not attached to seed portions of a plant. For example, trichome fibers, unlike seed hair fibers, are not attached to a seed or a seed pod epidermis. Cotton, kapok, milkweed, and coconut coir are non-limiting examples of seed hair fibers.

Further, trichome fibers are different from nonwood bast and/or core fibers in that they are not attached to the bast, also known as phloem, or the core, also known as xylem portions of a nonwood dicotyledonous plant stem. Non-limiting examples of plants which have been used to yield nonwood bast fibers and/or nonwood core fibers include kenaf, jute, flax, ramie and hemp.

Further trichome fibers are different from monocotyledonous plant derived fibers such as those derived from cereal straws (wheat, rye, barley, oat, etc), stalks (corn, cotton, sorghum, *Hesperaloe funifera*, etc.), canes (bamboo, bagasse, etc.), grasses (esparto, lemon, sabai, switchgrass, etc), since such monocotyledonous plant derived fibers are not attached to an epidermis of a plant.

Further, trichome fibers are different from leaf fibers in that they do not originate from within the leaf structure. Sisal and abaca are sometimes liberated as leaf fibers.

Finally, trichome fibers are different from wood pulp fibers since wood pulp fibers are not outgrowths from the epidermis of a plant; namely, a tree. Wood pulp fibers rather originate from the secondary xylem portion of the tree stem.

“Basis Weight” as used herein is the weight per unit area of a sample reported in lbs/3000 ft² or g/m² (gsm) and is measured according to the Basis Weight Test Method described herein.

“Machine Direction” or “MD” as used herein means the direction parallel to the flow of the fibrous structure through the fibrous structure making machine and/or sanitary tissue product manufacturing equipment.

“Cross Machine Direction” or “CD” as used herein means the direction parallel to the width of the fibrous structure making machine and/or sanitary tissue product manufacturing equipment and perpendicular to the machine direction.

“Embossed” as used herein with respect to an article, sanitary tissue product, and/or fibrous web (fibrous web ply), means that an article, sanitary tissue product, and/or fibrous web (fibrous web ply) has been subjected to a process which converts a smooth surfaced article, sanitary tissue product, and/or fibrous web (fibrous web ply) to an out-of-plane, textured surface by replicating a pattern on one or more emboss rolls, which form a nip through which the article, sanitary tissue product and/or fibrous web (fibrous web ply) passes. Embossed does not include creping, microcreping, printing or other processes that may also impart a texture and/or decorative pattern to an article, sanitary tissue product and/or fibrous web (fibrous web ply).

“Differential density”, as used herein, means a fibrous structure and/or fibrous web (fibrous web ply) that comprises one or more regions of relatively low fibrous element, for example fiber, density, which are referred to as pillow regions, and one or more regions of relatively high fibrous element, for example fiber, density, which are referred to as knuckle regions.

“Densified”, as used herein means a portion of a fibrous structure and/or fibrous web (fibrous web ply) that is characterized by regions of relatively high fibrous element, e.g., fiber, density (knuckle regions).

“Non-densified”, as used herein, means a portion of a fibrous structure and/or fibrous web (fibrous web ply) that exhibits a lesser fibrous element, e.g., fiber, density (one or more regions of relatively lower fibrous element, e.g., fiber, density) (pillow regions) than another portion (for example a knuckle region) of the fibrous structure and/or fibrous web (fibrous web ply).

“Wet textured” as used herein means that a three-dimensional (3D) patterned fibrous structure and/or 3D patterned fibrous web (3D patterned fibrous web ply) comprises texture (for example a three-dimensional topography) imparted to the fibrous structure and/or fibrous structure’s surface and/or fibrous web’s surface (fibrous web ply’s surface) during a fibrous structure making process. In one example, in a paper web, for example a fibrous structure making process, wet texture may be imparted to a fibrous structure upon fibers and/or filaments being collected on a collection device that has a three-dimensional (3D) surface which imparts a 3D surface to the fibrous structure being formed thereon and/or being transferred to a fabric and/or belt, such as a through-air-drying fabric and/or a patterned drying belt, comprising a 3D surface that imparts a 3D surface to a fibrous structure being formed thereon. In one example, the collection device with a 3D surface comprises a patterned, such as a patterned formed by a polymer or resin being deposited onto a base substrate, such as a fabric, in a patterned configuration. The wet texture imparted to a paper web, for example a fibrous structure is formed in the fibrous structure prior to and/or during drying of the fibrous structure. Non-limiting examples of collection devices and/or fabric and/or belts suitable for imparting wet texture to a fibrous structure include those fabrics and/or belts used in fabric creping and/or belt creping processes, for example as disclosed in U.S. Pat. Nos. 7,820,008 and 7,789,995, coarse through-air-drying fabrics as used in uncreped through-air-drying processes, and photo-curable resin patterned through-air-drying belts, for example as disclosed in U.S. Pat. No. 4,637,859. For purposes of the present invention, the collection devices used for imparting wet texture to the fibrous structures would be patterned to result in the fibrous structures comprising a surface pattern comprising a plurality of parallel line elements wherein at least one, two, three, or more, for example all of the parallel line elements exhibit a

non-constant width along the length of the parallel line elements. This is different from non-wet texture that is imparted to a fibrous structure after the fibrous structure has been dried, for example after the moisture level of the fibrous structure is less than 15% and/or less than 10% and/or less than 5%. An example of non-wet texture includes embossments imparted to a fibrous structure and/or fibrous web (fibrous web ply) by embossing rolls during converting of the fibrous structure and/or fibrous web (fibrous web ply). In one example, the fibrous structure and/or fibrous web (fibrous web ply), for example a paper web, for example a fibrous structure and/or wet-laid fibrous web (wet-laid fibrous web ply), is a wet textured fibrous structure and/or wet textured fibrous web (wet textured fibrous web ply).

"3D pattern" with respect to a fibrous structure and/or fibrous web's surface (fibrous web ply's surface) in accordance with the present invention means herein a pattern that is present on at least one surface of the fibrous structure and/or fibrous web (fibrous web ply). The 3D pattern texturizes the surface of the fibrous structure and/or fibrous web (fibrous web ply), for example by providing the surface with protrusions and/or depressions. The 3D pattern on the surface of the fibrous structure and/or fibrous web (fibrous web ply) is made by making the fibrous structure on a patterned molding member that imparts the 3D pattern to the fibrous structure made thereon. For example, the 3D pattern may comprise a series of line elements, such as a series of line elements that are substantially oriented in the cross-machine direction of the fibrous structure and/or sanitary tissue product.

In one example, a series of line elements may be arranged in a 3D pattern selected from the group consisting of: periodic patterns, aperiodic patterns, straight line patterns, curved line patterns, wavy line patterns, snaking patterns, square line patterns, triangular line patterns, S-wave patterns, sinusoidal line patterns, and mixtures thereof. In another example, a series of line elements may be arranged in a regular periodic pattern or an irregular periodic pattern (aperiodic) or a non-periodic pattern.

"Distinct from" and/or "different from" as used herein means two things that exhibit different properties and/or levels of materials, for example different by 0.5 and/or 1 and/or 2 and/or 3 and/or 5 and/or 10 units and/or different by 1% and/or 3% and/or 5% and/or 10% and/or 20%, different materials, and/or different average fiber diameters.

"Textured pattern" as used herein means a pattern, for example a surface pattern, such as a three-dimensional (3D) surface pattern present on a surface of the fibrous structure and/or on a surface of a component making up the fibrous structure.

"Fibrous Structure Basis Weight" as used herein is the weight per unit area of a sample reported in lbs/3000 ft² or g/m².

"Ply" as used herein means an individual, integral fibrous structure.

"Plies" as used herein means two or more individual, integral fibrous structures disposed in a substantially contiguous, face-to-face relationship with one another, forming a multi-ply sanitary tissue product. It is also contemplated that an individual, integral fibrous structure can effectively form a multi-ply sanitary tissue product, for example, by being folded on itself.

"Common Intensive Property" as used herein means an intensive property possessed by more than one region within a fibrous structure. Such intensive properties of the fibrous structure include, without limitation, density, basis weight, thickness, and combinations thereof. For example, if density

is a common intensive property of two or more different regions, a value of the density in one region can differ from a value of the density in one or more other regions. Regions (such as, for example, a first region and a second region and/or a continuous network region and at least one of a plurality of discrete zones) are identifiable areas visually discernible and/or visually distinguishable from one another by distinct intensive properties.

"X," "Y," and "Z" designate a conventional system of Cartesian coordinates, wherein mutually perpendicular coordinates "X" and "Y" define a reference X-Y plane, and "Z" defines an orthogonal to the X-Y plane. "Z-direction" designates any direction perpendicular to the X-Y plane. Analogously, the term "Z-dimension" means a dimension, distance, or parameter measured parallel to the Z-direction. When an element, such as, for example, a molding member curves or otherwise deplanes, the X-Y plane follows the configuration of the element.

"Substantially continuous" or "continuous" region refers to an area within which one can connect any two points by an uninterrupted line running entirely within that area throughout the line's length. That is, the substantially continuous region has a substantial "continuity" in all directions parallel to the first plane and is terminated only at edges of that region. The term "substantially," in conjunction with continuous, is intended to indicate that while an absolute continuity is preferred, minor deviations from the absolute continuity may be tolerable as long as those deviations do not appreciably affect the performance of the fibrous structure (or a molding member) as designed and intended.

"Substantially semi-continuous" or "semi-continuous" region refers to an area which has "continuity" in all, but at least one, directions parallel to the first plane, and in which area one cannot connect any two points by an uninterrupted line running entirely within that area throughout the line's length. The semi-continuous framework may have continuity only in one direction parallel to the first plane. By analogy with the continuous region, described above, while an absolute continuity in all, but at least one, directions is preferred, minor deviations from such a continuity may be tolerable as long as those deviations do not appreciably affect the performance of the fibrous structure.

"Discontinuous" or "discrete" regions or zones refer to discrete, and separated from one another areas or zones that are discontinuous in all directions parallel to the first plane.

"Molding member" is a structural element that can be used as a support for the mixture of filaments and solid additives that can be deposited thereon during a process of making a fibrous structure, and as a forming unit to form (or "mold") a desired microscopical geometry of a fibrous structure. The molding member may comprise any element that has the ability to impart a three-dimensional pattern to the fibrous structure being produced thereon, and includes, without limitation, a stationary plate, a belt, a cylinder/roll, a woven fabric, and a band.

As used herein, the articles "a" and "an" when used herein, for example, "an anionic surfactant" or "a fiber" is understood to mean one or more of the material that is claimed or described.

All percentages and ratios are calculated by weight unless otherwise indicated. All percentages and ratios are calculated based on the total composition unless otherwise indicated.

Unless otherwise noted, all component or composition levels are in reference to the active level of that component or composition, and are exclusive of impurities, for

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example, residual solvents or by-products, which may be present in commercially available sources.

Article

An article of the present invention comprises one or more and/or two or more and/or three or more and/or four or more fibrous webs (fibrous web plies), which comprise one or more fibrous structures, according to the present invention.

It has unexpectedly been found that the arrangement of the fibrous structures and/or fibrous webs (fibrous web plies) within the articles of the present invention and/or type of fibrous structures and/or type of fibrous elements, for example filaments and/or fibers, within the articles of the present invention result in the article of the present invention exhibiting novel properties, such as bulk and/or absorbent properties without negatively impacting the softness and/or flexibility and/or stiffness of the articles.

In one example, the articles of the present invention may comprise different combinations of fibrous webs (fibrous web plies) and/or fibrous structures and/or fibrous elements. For example, the articles of the present invention may comprise different combinations (associations) of wet-laid fibrous structures, for example 100% by weight of fibers, such as pulp fibers, for example wood pulp fibers (e.g., cellulosic wood pulp fibers) and co-formed fibrous structures, for example a mixture of filaments and fibers, such as polypropylene filaments and pulp fibers, such as wood pulp fibers (e.g., cellulosic wood pulp fibers), which allows for the creation of both wet and dry bulk, while maintaining a soft and/or flexibility and/or non-stiff sheet. This unique combination of properties is afforded, in this case, by the use of the co-formed fibrous structure, in which continuous filaments are combined with fibers in a way that the resultant bulk density of the sheet is very low. This low bulk density is maintained even when wet due the lack of collapse of the article, as the continuous filaments are not subject to water induced collapse. In contrast, such bulk in wet-laid fibrous structures is created via hydrogen bonding of the fibers within the wet-laid fibrous structure, which collapse if dry forming, such as embossing and/or microselfing, is used to create a soft fibrous structure with dry bulk (resulting in low wet bulk), or will be stiff if wet forming, such as forming the wet-laid fibrous structure on a molding member and/or subjecting the wet-laid fibrous structure to wet microcontraction during forming, is used to create a dry bulk that is resilient when wet.

In one example, the articles of the present invention comprise less than 50% and/or less than 40% and/or less than 30% and/or less than 25% and/or less than 20% and/or less than 15% and/or greater than 0% and/or greater than 5% by weight of filaments, for example thermoplastic filaments such as polyolefin filaments, for example polypropylene filaments.

In another example, the articles of the present invention allow for the optimization of different fibrous structures and/or fibrous webs (fibrous web plies) for different characteristics and/or properties. One example of this is how a very low density, high bulk co-formed fibrous structure that is strong can be placed with a wet formed, high bulk wet-laid fibrous structure that is very absorbent. The resultant article is one which is both highly absorbent, very compressible, and able to spring back after compression. This results in a spongelike article which is resilient under compression yet highly absorbent like a paper towel. Another example, of this is how a very low density, high bulk co-formed fibrous structure can be placed with a wet formed, high bulk wet-laid fibrous structure. The resultant article exhibits high bulk values when dry, are compressible under load and

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rebound when the load is relieved. Additionally, the resultant article exhibits high bulk, compressibility, and recovery when wet, due to the wet formed nature of the wet-laid fibrous structure and the co-formed fibrous structure, which is impervious to wet collapse.

In another example, the articles of the present invention exhibit very high sheet and/or roll bulk without negatively impacting softness. This high bulk can be achieved through multiple inner fibrous structures and/or fibrous webs (fibrous web plies), with the interior fibrous structures and/or fibrous webs (fibrous web plies) comprised of high loft, pin-holed wet-laid fibrous structures. Co-formed fibrous structures, which contain continuous, thermoplastic filaments and pulp fibers, enable the use of high loft wet-laid fibrous structures because the filaments are used for strength (especially when wet). Furthermore, the commingled nature of the filaments and fibers within the co-formed fibrous structures allows for very high bulk fibrous structures that are both absorbent and soft, as individual fibers are commingled within a network of continuous filaments. Articles like these are very difficult to make via other technologies such as solely wet-laid technology due to the fact that the fibers, such as pulp fibers, must impart strength and bulk and absorbency. These different demands in the past have caused product developers to optimize for some attributes at the expense of others.

In still another example, the articles of the present invention exhibit very high absorbencies without compromising softness of the article. This is achieved through the heterogeneous composition of the article; namely, the combination of at least two different fibrous structures, for example at least one co-formed fibrous structure and at least one wet-laid fibrous structure. To allow for high absorbencies, wet-laid fibrous structure making process choices such as fiber furnish mix, fiber refining levels, and molding member, for example belt design upon which the wet-laid fibrous structure is formed, can be chosen to create a lofty, high absorbent capacity wet-laid fibrous structure that is soft and low in strength. The filaments, for example polypropylene filaments, present in the co-formed fibrous structure is relied upon to deliver the strength of the article, while still being soft and/or flexible and/or non-stiff both wet and dry. Additionally, the interspersions of fibers, for example pulp fibers, with the filaments within the co-formed fibrous structure adds to the soft, velvet-like hand feel of the article.

In yet another example, the articles of the present invention exhibit very high absorbencies without compromising strength of the article. This is achieved through the heterogeneous composition of the article; namely, the combination of at least two different fibrous structures, for example at least one co-formed fibrous structure and at least one wet-laid fibrous structure. The wet-laid structure can be optimized for high absorbent capacities and/or rates without having to compromise to maintain strength. To allow for high absorbencies, wet-laid fibrous structure making process choices such as fiber furnish mix, fiber refining levels, and molding member, for example belt design upon which the wet-laid fibrous structure is formed, can be chosen to create a lofty, high absorbent capacity wet-laid fibrous structure that is soft and low in strength. The filaments, for example polypropylene filaments, present in the co-formed fibrous structure is relied upon to deliver the strength of the article, while still being soft and/or flexible and/or non-stiff both wet and dry. Additionally, the interspersions of fibers, for example pulp fibers, with the filaments within the co-formed fibrous structure adds to the soft, velvet-like hand feel of the article.

In another example, the articles of the present invention exhibit high absorbent capacity while still maintaining hand

protection. This can be achieved by tailoring the density, capillary pressure, and absorbent capacity of the different fibrous structures within the article. In one example, high density and capillary pressure wet-laid fibrous structures on one or both of the exterior surfaces of the article allow for rapid redistribution of water on a surface of the article, while lower density fibrous structure, such as co-formed fibrous structures, in the interior of the article creates storage capacity. In another example, thin, low density fibrous structures on one or more of the exterior surfaces of the article allow for rapid acquisition of water by the inner, more dense, high capillary pressure fibrous structures, such as wet-laid fibrous structures, whose high capillary pressure structures will redistribute the water in the article and not give it back to the exterior surfaces of the article.

In still another example, the articles of the present invention exhibit high bulk/low density without impacting the overall opacity of the articles. This can be achieved by the combining of differential density wet-laid fibrous structures, which have been wet formed such that relatively low density regions and relatively high density regions are formed in the wet-laid fibrous structure, to the extent that the low density regions of the wet-laid fibrous structure have very low basis weight, to the point of making pinholes. This is normally undesirable in wet-laid fibrous structures and/or wet-laid fibrous structure making processes, as the pinholes are detrimental to strength as well as opacity. When this wet-laid fibrous structure is combined with a co-formed fibrous structure the opacity significantly increases, creating a low density and high opacity article.

In yet another example, the articles of the present invention are very reopenable while still maintaining consumer acceptable absorbent properties. This is achieved through the combination of fibrous structures comprising filaments and/or a mixture of filaments and fibers, and wet-laid fibrous structures. In one example, low basis weight filament-containing fibrous structures, such as scrims of filaments, for example scrims of polypropylene filaments, are arranged on one or more of the exterior surfaces of the articles, which in turn further comprises one or more inner fibrous structures comprising wet-laid fibrous structures and co-formed fibrous structures. This combination of materials creates an article exhibits very high bulk absorbency and at the same time exhibits high wet resiliency, allowing it to be easily reopened during use, especially after being wetted.

In still another example, the articles of the present invention exhibit both high absorbent capacity and high surface drying properties. This combination is achieved through the combination of fibrous structures that exhibit different capillary pressures. One example of such an article that exhibits this characteristic is an article that has one or more wet-laid fibrous structure on one or more exterior surfaces of the articles, along with a co-formed fibrous structure as one or more inner fibrous structures within the articles. This low density co-formed fibrous structure core of the articles creates large absorbent capacity, while the wet-laid fibrous structure on the outside of the articles allows for consumer acceptable surface drying.

In even yet another example, the articles of the present invention exhibit both high wet bulk and high surface drying properties. This combination is achieved through the combination of fibrous structures that exhibit high capillary pressure with fibrous structures that exhibit high bulk when wet. One example of such an article that exhibits these characteristic is one that has one or more wet-laid fibrous structures on one or more exterior surfaces of an article, along with a co-formed fibrous structure in the center of the

article. The co-formed fibrous structure core does not collapse when wetted, while the wet-laid fibrous structure on the outside of the article allows for consumer acceptable surface drying.

Non-limiting examples of articles of the present invention are described below in more detail.

In one example, as shown in FIG. 3, an article **20** of the present invention comprises three fibrous webs (fibrous web plies): 1) a first fibrous web (fibrous web ply) example of which is shown in FIGS. **2A** and **2B** comprising a co-formed fibrous structure **22** (a multi-fibrous element fibrous structure) associated with two meltblown fibrous structures **24** (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure **22**, 2) a second fibrous web (fibrous web ply) example of which is shown in FIGS. **2A** and **2B** comprising a co-formed fibrous structure **22** (a multi-fibrous element fibrous structure) associated with two meltblown fibrous structures **24**, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure **22**, and 3) a third fibrous web (fibrous web ply) comprising a paper web, for example a fibrous structure **26** (a mono-fibrous element fibrous structure), for example a textured fibrous structure, for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure, positioned between and associated with at least one and/or both of the first and second fibrous webs, the co-formed fibrous webs **28** (co-formed fibrous web plies). The fibrous webs may be associated with each other in one operation or in multiple operations, such as by combining two of the fibrous webs first and then combining the remaining fibrous web with the already combined fibrous webs. In one example, the article **20** shown in FIG. **3** is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIG. **4**, an article **20** of the present invention comprises four fibrous webs (fibrous web plies) similar to the article shown in FIG. **3** above: 1) a first fibrous web (fibrous web ply) example of which is shown in FIGS. **2A** and **2B** comprising a co-formed fibrous structure **22** (a multi-fibrous element fibrous structure) associated with two meltblown fibrous structures **24**, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure **22**, 2) a second fibrous web (fibrous web ply) example of which is shown in FIGS. **2A** and **2B** comprising a co-formed fibrous structure **22** (a multi-fibrous element fibrous structure) associated with two meltblown fibrous structures **24** (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure, and 3) third and fourth fibrous webs (fibrous web plies) comprising paper webs, for example wet-laid fibrous structures **26**, (mono-fibrous element fibrous structures), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure, positioned between and associated with at least one and/or both of the first and second fibrous webs. The fibrous webs may be associated with each other in one operation or in multiple operations, such as by combining two or three of the fibrous webs first and then combining the remaining fibrous webs with the already combined fibrous webs. In one example, the article **20** shown in FIG. **4** is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIG. **5**, an article **20** of the present invention comprises two fibrous webs (fibrous web plies): 1) a fibrous web (fibrous web ply) example of which

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is shown in FIGS. 2A and 2B comprising a co-formed fibrous structure 22 (multi-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure 22, and 2) a second fibrous web (fibrous web ply) example of which is shown in FIGS. 6A and 6B comprising a co-formed fibrous structure 22 (multi-fibrous element fibrous structure) associated with one meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on one surface of the co-formed fibrous structure 22 and a paper web, for example a wet-laid fibrous structure 26 (a mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure on the opposite surface of the co-formed fibrous structure 22. The paper web, for example the wet-laid fibrous structure 26 may be further associated with a meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on the wet-laid fibrous structure's surface opposite the co-formed fibrous structure 22. The fibrous webs may be associated with each other in one operation, such as by combining the two fibrous webs such that the paper web, for example the wet-laid fibrous structure 26 is positioned between the two co-formed fibrous structures 22 in the article 20. In one example, the article 20 shown in FIG. 5 is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIG. 7, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) two fibrous webs (fibrous web plies) examples of which are shown in FIGS. 6A and 6B comprising a co-formed fibrous structure 22 (multi-fibrous element fibrous structure) associated with one meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on one surface of the co-formed fibrous structure 22 and a paper web, for example a wet-laid fibrous structure 26 (a mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure on the opposite surface of the fibrous structure. The paper web, for example the wet-laid fibrous structure 26 may be further associated with a meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on the wet-laid fibrous structure's surface opposite the co-formed fibrous structure 22. The fibrous webs may be associated with each other in one operation, such as by combining the two fibrous webs such that the paper webs, for example the wet-laid fibrous structures 26 are positioned between the two co-formed fibrous structures 22 in the article 20. In one example, the article 20 shown in FIG. 7 is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIG. 8, an article 20 of the present invention comprises a single fibrous web (fibrous web ply): 1) a fibrous web (fibrous web ply) example of which is shown in FIGS. 9A and 9B comprising a paper web, for example a wet-laid fibrous structure 26, such as a textured fibrous structure, (mono-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the wet-laid fibrous structure 26.

In one example, as shown in FIG. 10, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) two fibrous webs (fibrous web plies) examples of

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which are shown in FIGS. 9A and 9B comprising a paper web, for example a wet-laid fibrous structure 26, such as a textured fibrous structure, (mono-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the paper web, for example the wet-laid fibrous structure 26. In one example, the article 20 shown in FIG. 10 is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIG. 11, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) a first fibrous web (fibrous web ply) example of which is shown in FIGS. 9A and 9B comprising a paper web, for example a wet-laid fibrous structure 26, such as a textured fibrous structure, (mono-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the wet-laid fibrous structure 26, and 2) a second fibrous web (fibrous web ply) example of which is shown in FIGS. 6A and 6B comprising a co-formed fibrous structure 22 (multi-fibrous element fibrous structure) associated with one meltblown fibrous structure 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structure) on one surface of the co-formed fibrous structure 22 and a paper web, for example a wet-laid fibrous structure 26 (a mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure on the opposite surface of the fibrous structure. The paper web, for example the wet-laid fibrous structure 26 may be further associated with a meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on the wet-laid fibrous structure's surface opposite the co-formed fibrous structure 22. The fibrous webs may be associated with each other in one operation, such as by combining the two fibrous webs such that the paper webs, for example the wet-laid fibrous structures 26 are positioned as shown in FIG. 11. In one example, the article 20 shown in FIG. 11 is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIG. 12, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) a first fibrous web (fibrous web ply) example of which is shown in FIGS. 9A and 9B comprising a paper web, for example a wet-laid fibrous structure 26, such as a textured fibrous structure, (mono-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the wet-laid fibrous structure 26, and 2) a second fibrous web (fibrous web ply) example of which is shown in FIGS. 2A and 2B comprising a co-formed fibrous structure 22 (multi-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure 22. The fibrous webs may be associated with each other in one operation, such as by combining the two fibrous webs as shown in FIG. 12. In one example, the article 20 shown in FIG. 12 is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIG. 13, an article 20 of the present invention comprises a single fibrous web (fibrous web ply): 1) a fibrous web (fibrous web ply) example of

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which is shown in FIGS. 14A and 14B comprising a co-formed fibrous structure 22 (multi-fibrous element fibrous structure) associated with one meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on one surface of the co-formed fibrous structure 22 and a paper web, for example a wet-laid fibrous structure 26 (a mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure on the opposite surface of the co-formed fibrous structure 22. The paper web, for example the wet-laid fibrous structure 26 may be further associated with another co-formed fibrous structure 22 which in turn may be associated with another meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) such that the paper web, for example the wet-laid fibrous structure 26 is positioned between the two co-formed fibrous structures 22.

In one example, as shown in FIG. 15, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) two fibrous webs (fibrous web plies) examples of which are shown in FIGS. 6A and 6B comprising a two different co-formed fibrous structures 22 or a variably density (in the z-direction) co-formed fibrous structure 28 example of which is shown in FIGS. 16A and 16B (multi-fibrous element fibrous structure) associated with one meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on one surface of the co-formed fibrous structure 22 and a paper web, for example a wet-laid fibrous structure 26 (a mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure on the opposite surface of the fibrous structure. The paper web, for example the wet-laid fibrous structure 26 may be further associated with a meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on the wet-laid fibrous structure's surface opposite the co-formed fibrous structure 22. The fibrous webs may be associated with each other in one operation, such as by combining the two fibrous webs such that the paper webs, for example the wet-laid fibrous structures 26 are positioned between the two co-formed fibrous structures 22 in the article 20. In one example, the article 20 shown in FIG. 15 is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIG. 17, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) two fibrous webs (fibrous web plies) examples of which are shown in FIGS. 6A and 6B comprising a co-formed fibrous structure 22 (multi-fibrous element fibrous structure) associated with one meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on one surface of the co-formed fibrous structure 22 and a paper web, for example a wet-laid fibrous structure 26 (a mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure on the opposite surface of the fibrous structure. The paper web, for example the wet-laid fibrous structure 26 may be further associated with a meltblown fibrous structure 24, for example a scrim layer of filaments, (mono-fibrous element fibrous structure) on the wet-laid fibrous structure's surface opposite the co-formed fibrous structure 22. The fibrous webs may be associated with each other in one operation, such as by combining the two fibrous webs such that the co-formed fibrous structures 22 are positioned between the two paper webs, for example the two wet-laid fibrous structures 26 in

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the article 20. In one example, the article 20 shown in FIG. 17 is made by combining the pre-formed fibrous webs (fibrous web plies). The article 20 shown in FIG. 17 is similar to the article 20 shown in FIG. 7, with a different arrangement of the fibrous webs within the article 20.

In one example, as shown in FIG. 18, an article 20 of the present invention comprises three fibrous webs (fibrous web plies): 1) a first fibrous web (fibrous web ply) example of which is shown in FIGS. 2A and 2B comprising a co-formed fibrous structure 22 (a multi-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure 22 forming a co-formed fibrous web 28, 2) second and third fibrous webs (fibrous web plies) comprising paper webs, for example wet-laid fibrous structures 26 (mono-fibrous element fibrous structures), for example a textured fibrous structure, for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure associated with the co-formed fibrous web 28 (co-formed fibrous web plies). The paper webs, for example the wet-laid fibrous structure 26 may also be associated with one or more meltblown fibrous structures 24, for example one or more scrim layers of filaments, present on one or both of the wet-laid fibrous structure's surfaces. FIG. 19 shows a similar article 20 to that shown in FIG. 18 except that the paper web, for example the wet-laid fibrous structure 26 forms at least one or both of the exterior surfaces of the article 20. In other words, the paper web, for example the wet-laid fibrous structure 26 is not associated with a meltblown fibrous structure 24, for example not associated with a scrim layer of filaments, that forms an exterior surface of the article 20. The fibrous webs may be associated with each other in one operation or in multiple operations, such as by combining two of the fibrous webs first and then combining the remaining fibrous web with the already combined fibrous webs. In one example, the article 20 shown in FIG. 18 is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIG. 20, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) two fibrous webs (fibrous web plies) examples of which are shown in FIGS. 21A and 21B comprising a co-formed fibrous structure 22 (a multi-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure 22 forming a co-formed fibrous web 28, wherein the co-formed fibrous web 28 is associated with a paper web, for example a wet-laid fibrous structure 26 (mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure. The combined webs may be embossed in an emboss nip 33 formed by one or more patterned emboss rolls 39, one or more of which may be heated. The paper web, for example the wet-laid fibrous structure 26 may be associated with one or more meltblown fibrous structures 24, for example one or more scrim layers of filaments, present on one or both of the wet-laid fibrous structure's surfaces. The fibrous webs may be associated with each other in one operation, such as by combining the fibrous webs (fibrous web plies) such that the paper webs, for example the wet-laid fibrous structures 26 are positioned between the co-formed fibrous webs 28. In one example, the article 20 shown in FIG. 20 is made by combining the pre-formed fibrous webs (fibrous web plies).

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In one example, as shown in FIGS. 22A and 22B, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) two fibrous webs (fibrous web plies) examples of which are shown in FIGS. 23A and 23B comprising a co-formed fibrous structure 22 (a multi-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure 22 forming a co-formed fibrous web 28, wherein the co-formed fibrous web 28 is associated with a paper web, for example a wet-laid fibrous structure 26 (mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure. The paper webs, for example the wet-laid fibrous structures 26 may be formed on a textured collection device 31 and passed through a nip 33 formed between two rolls 41, for example a heated steel roll and a rubber roll. The paper web, for example the wet-laid fibrous structure 26 may be associated with one or more meltblown fibrous structures 24, for example one or more scrim layers of filaments, present on one or both of the wet-laid fibrous structure's surfaces. The fibrous webs may be associated with each other in one operation, such as by combining the fibrous webs (fibrous web plies) such that the paper webs, for example the wet-laid fibrous structures 26 are positioned between the co-formed fibrous webs 28. In one example, the article 20 shown in FIGS. 22A and 22B is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIGS. 24A and 24B, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) two fibrous webs (fibrous web plies) examples of which are shown in FIGS. 25A and 25B comprising a co-formed fibrous structure 22 (a multi-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed fibrous structure 22 forming a co-formed fibrous web 28, wherein the co-formed fibrous web 28 is associated with a paper web, for example a wet-laid fibrous structure 26 (mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure. The paper webs, for example the wet-laid fibrous structures 26 may be formed on a textured collection device 31 and passed through a nip 33 formed between two rolls 41, for example a heated steel roll and a rubber roll. The paper web, for example the wet-laid fibrous structure 26 may be associated with one or more meltblown fibrous structures 24, for example one or more scrim layers of filaments, present on one or both of the wet-laid fibrous structure's surfaces. The fibrous webs may be associated with each other in one operation, such as by combining the fibrous webs (fibrous web plies) such that the paper webs, for example the wet-laid fibrous structures 26 are positioned between the co-formed fibrous webs 28. In one example, the article 20 shown in FIGS. 24A and 24B is made by combining the pre-formed fibrous webs (fibrous web plies).

In one example, as shown in FIGS. 26A and 26B, an article 20 of the present invention comprises two fibrous webs (fibrous web plies): 1) two fibrous webs (fibrous web plies) examples of which are shown in FIGS. 27A and 27B comprising a co-formed fibrous structure 22 (a multi-fibrous element fibrous structure) associated with two meltblown fibrous structures 24, for example two scrim layers of filaments, (mono-fibrous element fibrous structures), which function as scrims on opposite surfaces of the co-formed

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fibrous structure 22 forming a co-formed fibrous web 28, wherein the co-formed fibrous web 28 is associated with a paper web, for example a wet-laid fibrous structure 26 (mono-fibrous element fibrous structure), for example a textured wet-laid fibrous structure, such as a 3D patterned wet-laid fibrous structure. The combined webs may be embossed in an emboss nip 33 formed by one or more patterned emboss rolls 39, one or more of which may be heated. The paper web, for example the wet-laid fibrous structure 26 may be associated with one or more meltblown fibrous structures 24, for example one or more scrim layers of filaments, present on one or both of the wet-laid fibrous structure's surfaces. The fibrous webs may be associated with each other in one operation, such as by combining the fibrous webs (fibrous web plies) such that the paper webs, for example the wet-laid fibrous structures 26 are positioned between the co-formed fibrous webs 28. In one example, the article 20 shown in FIGS. 26A and 26B is made by combining the pre-formed fibrous webs (fibrous web plies).

Any of the meltblown fibrous structures 24 may be optional, especially if they represent an exterior surface of the articles 20. In one example, the article 20 of FIG. 11 may be void of the meltblown fibrous structure 24 forming the exterior surface of the article 20, which is associated with the paper web, for example the wet-laid fibrous structure 26.

In another example, the combined fibrous webs shown in FIG. 23A may be combined with a paper web, for example a wet-laid fibrous structure 26 to form an article 20. The paper web, for example the wet-laid fibrous structure 26 may be void of a meltblown fibrous structure 24 or may comprise one or more, two or more, meltblown fibrous structures 24 on at least one exterior surface and/or on both exterior surfaces (opposite surfaces).

The articles of the present invention and/or any fibrous webs of the present invention may be subjected to any post-processing operations such as embossing operations, printing operations, tuft-generating operations, thermal bonding operations, ultrasonic bonding operations, perforating operations, surface treatment operations such as application of lotions, silicones and/or other materials and mixtures thereof.

Fibrous Webs (Fibrous Web Plies)

Non-limiting examples of fibrous webs (fibrous web plies) according to the present invention comprise one or more and/or two or more and/or three or more and/or four or more and/or five or more and/or six or more and/or seven or more fibrous structures that are associated with one another, such as by compression bonding (for example by passing through a nip formed by two rollers), thermal bonding (for example by passing through a nip formed by two rollers where at least one of the rollers is heated to a temperature of at least about 120° C. (250° F.)), microselfing, needle punching, and gear rolling, to form a unitary structure.

Wet-Laid Fibrous Structure (an example of a Mono-Fibrous Element Fibrous Structure)

The wet-laid fibrous structure comprises a plurality of fibrous elements, for example a plurality of fibers. In one example, the wet-laid fibrous structure comprises a plurality of naturally-occurring fibers, for example pulp fibers, such as wood pulp fibers (hardwood and/or softwood pulp fibers). In another example, the wet-laid fibrous structure comprises a plurality of non-naturally occurring fibers (synthetic fibers), for example staple fibers, such as rayon, lyocell, polyester fibers, polycaprolactone fibers, polylactic acid fibers, polyhydroxyalkanoate fibers, and mixtures thereof.

The mono-fibrous element fibrous structure may comprise one or more filaments, such as polyolefin filaments, for

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example polypropylene and/or polyethylene filaments, starch filaments, starch derivative filaments, cellulose filaments, polyvinyl alcohol filaments.

The wet-laid fibrous structure of the present invention may be single-ply or multi-ply web material. In other words, the wet-laid fibrous structures of the present invention may comprise one or more wet-laid fibrous structures, the same or different from each other so long as one of them comprises a plurality of pulp fibers.

In one example, the wet-laid fibrous structure comprises a wet laid fibrous structure ply, such as a through-air-dried fibrous structure ply, for example an uncreped, through-air-dried fibrous structure ply and/or a creped, through-air-dried fibrous structure ply.

In another example, the wet-laid fibrous structure and/or wet laid fibrous structure ply may exhibit substantially uniform density.

In another example, the wet-laid fibrous structure and/or wet laid fibrous structure ply may comprise a surface pattern. In one example, the surface pattern comprises a one or more relatively high density regions and one or more relatively low density regions. In another example, the surface pattern comprises one or more relatively high elevation regions and one or more relatively low elevation regions. In yet another example, the surface pattern comprises one or more relatively high basis weight regions and one or more relatively low basis weight regions. In still another example, the surface pattern is a non-random, repeating pattern, which may comprise a plurality of discrete regions dispersed throughout a continuous network. At least a portion of the plurality of discrete regions may exhibit a value of a common intensive property (such as density, bulk, and/or basis weight) that is different from the value of the common intensive property exhibited by the continuous network.

In one example, the wet laid fibrous structure ply comprises a conventional wet-pressed fibrous structure ply. The wet laid fibrous structure ply may comprise a fabric-creped fibrous structure ply. The wet laid fibrous structure ply may comprise a belt-creped fibrous structure ply.

In still another example, the wet-laid fibrous structure may comprise an air laid fibrous structure ply.

The wet-laid fibrous structures of the present invention may comprise a surface softening agent or be void of a surface softening agent, such as silicones, quaternary ammonium compounds, lotions, and mixtures thereof. In one example, the sanitary tissue product is a non-lotioned wet-laid fibrous structure.

The wet-laid fibrous structures of the present invention may comprise trichome fibers or may be void of trichome fibers.

Patterned Molding Members

The wet-laid fibrous structures of the present invention may be formed on patterned molding members that result in the wet-laid fibrous structures of the present invention. In one example, the pattern molding member comprises a non-random repeating pattern. In another example, the pattern molding member comprises a resinous pattern.

In one example, the wet-laid fibrous structure comprises a textured surface. In another example, the wet-laid fibrous structure comprises a surface comprising a three-dimensional (3D) pattern, for example a 3D pattern imparted to the wet-laid fibrous structure by a patterned molding member. Non-limiting examples of suitable patterned molding members include patterned felts, patterned forming wires, patterned rolls, patterned fabrics, and patterned belts utilized in conventional wet-pressed papermaking processes, air-laid

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papermaking processes, and/or wet-laid papermaking processes that produce 3D patterned sanitary tissue products and/or 3D patterned fibrous structure plies employed in sanitary tissue products. Other non-limiting examples of such patterned molding members include through-air-drying fabrics and through-air-drying belts utilized in through-air-drying papermaking processes that produce through-air-dried fibrous structures, for example 3D patterned through-air dried fibrous structures, and/or through-air-dried sanitary tissue products comprising the wet-laid fibrous structure.

A "reinforcing element" may be a desirable (but not necessary) element in some examples of the molding member, serving primarily to provide or facilitate integrity, stability, and durability of the molding member comprising, for example, a resinous material. The reinforcing element can be fluid-permeable or partially fluid-permeable, may have a variety of embodiments and weave patterns, and may comprise a variety of materials, such as, for example, a plurality of interwoven yarns (including Jacquard-type and the like woven patterns), a felt, a plastic, other suitable synthetic material, or any combination thereof.

Non-limiting examples of patterned molding members suitable for use in the present invention comprises a through-air-drying belts. The through-air-drying belts may comprise a plurality of continuous knuckles, discrete knuckles, semi-continuous knuckles and/or continuous pillows, discrete pillows, and semi-continuous pillows formed by resin arranged in a non-random, repeating pattern supported on a support fabric comprising filaments, such as a forming fabric. The resin is patterned such that deflection conduits that contain little to no resin present in the pattern and result in the fibrous structure being formed on the patterned molding member having one or more pillow regions (low density regions) compared to the knuckle regions that are imparted to the fibrous structure by the resin areas.

Non-Limiting Examples of Making Wet-Laid Fibrous Structures

In one non-limiting example, the wet-laid fibrous structure is made on a molding member of the present invention. The method may be a paper web, for example a fibrous structure making process that uses a cylindrical dryer such as a Yankee (a Yankee-process) (creped) or it may be a Yankeeless process (uncreped) as is used to make substantially uniform density and/or uncreped wet-laid fibrous structures (fibrous structures).

In one example, a process for making a paper web, for example a fibrous structure according to the present invention comprises supplying an aqueous dispersion of fibers (a fibrous or fiber furnish or fiber slurry) to a headbox which can be of any convenient design. From the headbox the aqueous dispersion of fibers is delivered to a first foraminous member (forming wire) which is typically a Fourdrinier wire, to produce an embryonic fibrous structure.

The embryonic fibrous structure is brought into contact with a patterned molding member, such as a 3D patterned through-air-drying belt. While in contact with the patterned molding member, the embryonic fibrous structure will be deflected, rearranged, and/or further dewatered. This can be accomplished by applying differential speeds and/or pressures.

After the embryonic fibrous structure has been associated with the patterned molding member, fibers within the embryonic fibrous structure are deflected into pillows ("deflection conduits") present in the patterned molding member. In one example of this process step, there is essentially no water removal from the embryonic fibrous structure through the deflection conduits after the embryonic fibrous structure

has been associated with the patterned molding member but prior to the deflecting of the fibers into the deflection conduits. Further water removal from the embryonic fibrous structure can occur during and/or after the time the fibers are being deflected into the deflection conduits. Water removal from the embryonic fibrous structure may continue until the consistency of the embryonic fibrous structure associated with patterned molding member is increased to from about 25% to about 35%. Once this consistency of the embryonic fibrous structure is achieved, then the embryonic fibrous structure can be referred to as an intermediate fibrous structure. As noted, water removal occurs both during and after deflection; this water removal may result in a decrease in fiber mobility in the embryonic web material. This decrease in fiber mobility may tend to fix and/or freeze the fibers in place after they have been deflected and rearranged. Of course, the drying of the web material in a later step in the process of this invention serves to more firmly fix and/or freeze the fibers in position.

Any convenient means conventionally known in the papermaking art can be used to dry the intermediate fibrous structure. Examples of such suitable drying process include subjecting the intermediate fibrous structure to conventional and/or flow-through dryers and/or Yankee dryers.

In one example of a drying process, the intermediate fibrous structure may first pass through an optional predryer. This predryer can be a conventional flow-through dryer (hot air dryer) well known to those skilled in the art. Optionally, the predryer can be a so-called capillary dewatering apparatus. In such an apparatus, the intermediate fibrous structure passes over a sector of a cylinder having preferential-capillary-size pores through its cylindrical-shaped porous cover. Optionally, the predryer can be a combination capillary dewatering apparatus and flow-through dryer. The quantity of water removed in the predryer may be controlled so that a predried fibrous structure exiting the predryer has a consistency of from about 30% to about 98%. The predried fibrous structure may be applied to a surface of a Yankee dryer via a nip with pressure, the pattern formed by the top surface of patterned molding member is impressed into the predried web material to form a 3D patterned fibrous structure, for example a 3D patterned wet-laid fibrous structure of the present invention. The 3D patterned wet-laid fibrous structure is then adhered to the surface of the Yankee dryer where it can be dried to a consistency of at least about 95%.

The 3D patterned wet-laid fibrous structure can then be foreshortened by creping the 3D patterned wet-laid fibrous structure with a creping blade to remove the 3D patterned wet-laid fibrous structure from the surface of the Yankee dryer resulting in the production of a 3D patterned creped wet-laid fibrous structure in accordance with the present invention. As used herein, foreshortening refers to the reduction in length of a dry (having a consistency of at least about 90% and/or at least about 95%) web material which occurs when energy is applied to the dry web material in such a way that the length of the dry web material is reduced and the fibers in the dry web material are rearranged with an accompanying disruption of fiber-fiber bonds. Foreshortening can be accomplished in any of several well-known ways. One common method of foreshortening is creping. Another method of foreshortening that is used to make the wet-laid fibrous structures of the present invention is wet microcontraction. Further, the wet-laid fibrous structure may be subjected to post processing steps such as calendaring, tuft generating operations, and/or embossing and/or converting.

Co-formed Fibrous Structures

The co-formed fibrous structures of the present invention comprise a plurality of filaments and a plurality of solid additives. The filaments and the solid additives may be commingled together. In one example, the fibrous structure is a coform fibrous structure comprising filaments and solid additives. The filaments may be present in the fibrous structures of the present invention at a level of less than 90% and/or less than 80% and/or less than 65% and/or less than 50% and/or greater than 5% and/or greater than 10% and/or greater than 20% and/or from about 10% to about 50% and/or from about 25% to about 45% by weight of the fibrous structure on a dry basis.

The solid additives may be present in the fibrous structures of the present invention at a level of greater than 10% and/or greater than 25% and/or greater than 50% and/or less than 100% and/or less than 95% and/or less than 90% and/or less than 85% and/or from about 30% to about 95% and/or from about 50% to about 85% by weight of the fibrous structure on a dry basis.

The filaments and solid additives may be present in the fibrous structures of the present invention at a weight ratio of filaments to solid additive of greater than 10:90 and/or greater than 20:80 and/or less than 90:10 and/or less than 80:20 and/or from about 25:75 to about 50:50 and/or from about 30:70 to about 45:55. In one example, the filaments and solid additives are present in the fibrous structures of the present invention at a weight ratio of filaments to solid additives of greater than 0 but less than 1.

In one example, the fibrous structures of the present invention exhibit a basis weight of from about 10 gsm to about 1000 gsm and/or from about 10 gsm to about 500 gsm and/or from about 15 gsm to about 400 gsm and/or from about 15 gsm to about 300 gsm as measured according to the Basis Weight Test Method described herein. In another example, the fibrous structures of the present invention exhibit a basis weight of from about 10 gsm to about 200 gsm and/or from about 20 gsm to about 150 gsm and/or from about 25 gsm to about 125 gsm and/or from about 30 gsm to about 100 gsm and/or from about 30 gsm to about 80 gsm as measured according to the Basis Weight Test Method described herein. In still another example, the fibrous structures of the present invention exhibit a basis weight of from about 80 gsm to about 1000 gsm and/or from about 125 gsm to about 800 gsm and/or from about 150 gsm to about 500 gsm and/or from about 150 gsm to about 300 gsm as measured according to the Basis Weight Test Method described herein.

In one example, the fibrous structure of the present invention comprises a core component. A "core component" as used herein means a fibrous structure comprising a plurality of filaments and optionally a plurality of solid additives. In one example, the core component is a coform fibrous structure comprising a plurality of filaments and a plurality of solid additives, for example pulp fibers. In one example, the core component is the component that exhibits the greatest basis weight with the fibrous structure of the present invention. In one example, the total core components present in the fibrous structures of the present invention exhibit a basis weight that is greater than 50% and/or greater than 55% and/or greater than 60% and/or greater than 65% and/or greater than 70% and/or less than 100% and/or less than 95% and/or less than 90% of the total basis weight of the fibrous structure of the present invention as measured according to the Basis Weight Test Method described herein. In another example, the core component exhibits a basis weight of greater than 12 gsm and/or greater than 14 gsm and/or greater than 16 gsm and/or greater than 18 gsm and/or

greater than 20 gsm and/or greater than 25 gsm as measured according to the Basis Weight Test Method described herein.

"Consolidated region" as used herein means a region within a fibrous structure where the filaments and optionally the solid additives have been compressed, compacted, and/or packed together with pressure and optionally heat (greater than 150° F.) to strengthen the region compared to the same region in its unconsolidated state or a separate region which did not see the compression or compacting pressure. In one example, a region is consolidated by forming unconsolidated regions within a fibrous structure on a patterned molding member and passing the unconsolidated regions within the fibrous structure while on the patterned molding member through a pressure nip, such as a heated metal anvil roll (about 275° F.) and a rubber anvil roll with pressure to compress the unconsolidated regions into one or more consolidated regions. In one example, the filaments present in the consolidated region, for example on the side of the fibrous structure that is contacted by the heated roll comprises fused filaments that create a skin on the surface of the fibrous structure, which may be visible via SEM images.

The fibrous structure of the present invention may, in addition a core component, further comprise a scrim component. "Scrim component" as used herein means a fibrous structure comprising a plurality of filaments. In one example, the total scrim components present in the fibrous structures of the present invention exhibit a basis weight that is less than 25% and/or less than 20% and/or less than 15% and/or less than 10% and/or less than 7% and/or less than 5% and/or greater than 0% and/or greater than 1% of the total basis weight of the fibrous structure of the present invention as measured according to the Basis Weight Test Method described herein. In another example, the scrim component exhibits a basis weight of 10 gsm or less and/or less than 10 gsm and/or less than 8 gsm and/or less than 6 gsm and/or greater than 5 gsm and/or less than 4 gsm and/or greater than 0 gsm and/or greater than 1 gsm as measured according to the Basis Weight Test Method described herein.

In one example, at least one of the core components of the fibrous structure comprises a plurality of solid additives, for example pulp fibers, such as comprise wood pulp fibers and/or non-wood pulp fibers.

In one example, at least one of the core components of the fibrous structure comprises a plurality of core filaments. In another example, at least one of the core components comprises a plurality of solid additives and a plurality of the core filaments. In one example, the solid additives and the core filaments are present in a layered orientation within the core component. In one example, the core filaments are present as a layer between two solid additive layers. In another example, the solid additives and the core filaments are present in a coform layer. At least one of the core filaments comprises a polymer, for example a thermoplastic polymer, such as a polyolefin. The polyolefin may be selected from the group consisting of: polypropylene, polyethylene, and mixtures thereof. In another example, the thermoplastic polymer of the core filament may comprise a polyester.

In one example, at least one of the scrim components is adjacent to at least one of the core components within the fibrous structure. In another example, at least one of the core components is positioned between two scrim components within the fibrous structure.

In one example, at least one of the scrim components of the fibrous structure of the present invention comprises a plurality of scrim filaments, for example scrim filaments, wherein the scrim filaments comprise a polymer, for

example a thermoplastic and/or hydroxyl polymer as described above with reference to the core components.

In one example, at least one of the scrim filaments exhibits an average fiber diameter of less than 50 and/or less than 25 and/or less than 10 and/or at least 1 and/or greater than 1 and/or greater than 3 μm as measured according to the Average Diameter Test Method described herein.

The average fiber diameter of the core filaments is less than 250 and/or less than 200 and/or less than 150 and/or less than 100 and/or less than 50 and/or less than 30 and/or less than 25 and/or less than 10 and/or greater than 1 and/or greater than 3 μm as measured according to the Average Diameter Test Method described herein.

In one example, the fibrous structures of the present invention may comprise any suitable amount of filaments and any suitable amount of solid additives. For example, the fibrous structures may comprise from about 10% to about 70% and/or from about 20% to about 60% and/or from about 30% to about 50% by dry weight of the fibrous structure of filaments and from about 90% to about 30% and/or from about 80% to about 40% and/or from about 70% to about 50% by dry weight of the fibrous structure of solid additives, such as wood pulp fibers.

In one example, the filaments and solid additives of the present invention may be present in fibrous structures according to the present invention at weight ratios of filaments to solid additives of from at least about 1:1 and/or at least about 1:1.5 and/or at least about 1:2 and/or at least about 1:2.5 and/or at least about 1:3 and/or at least about 1:4 and/or at least about 1:5 and/or at least about 1:7 and/or at least about 1:10.

In one example, the solid additives, for example wood pulp fibers, may be selected from the group consisting of softwood kraft pulp fibers, hardwood pulp fibers, and mixtures thereof. Non-limiting examples of hardwood pulp fibers include fibers derived from a fiber source selected from the group consisting of: *Acacia*, *Eucalyptus*, Maple, Oak, Aspen, Birch, Cottonwood, Alder, Ash, Cherry, Elm, Hickory, Poplar, Gum, Walnut, Locust, Sycamore, Beech, *Catalpa*, *Sassafras*, *Gmelina*, *Albizia*, *Anthocephalus*, and *Magnolia*. Non-limiting examples of softwood pulp fibers include fibers derived from a fiber source selected from the group consisting of: Pine, Spruce, Fir, Tamarack, Hemlock, Cypress, and Cedar. In one example, the hardwood pulp fibers comprise tropical hardwood pulp fibers. Non-limiting examples of suitable tropical hardwood pulp fibers include *Eucalyptus* pulp fibers, *Acacia* pulp fibers, and mixtures thereof.

In one example, the wood pulp fibers comprise softwood pulp fibers derived from the kraft process and originating from southern climates, such as Southern Softwood Kraft (SSK) pulp fibers. In another example, the wood pulp fibers comprise softwood pulp fibers derived from the kraft process and originating from northern climates, such as Northern Softwood Kraft (NSK) pulp fibers.

The wood pulp fibers present in the fibrous structure may be present at a weight ratio of softwood pulp fibers to hardwood pulp fibers of from 100:0 and/or from 90:10 and/or from 86:14 and/or from 80:20 and/or from 75:25 and/or from 70:30 and/or from 60:40 and/or about 50:50 and/or to 0:100 and/or to 10:90 and/or to 14:86 and/or to 20:80 and/or to 25:75 and/or to 30:70 and/or to 40:60. In one example, the weight ratio of softwood pulp fibers to hardwood pulp fibers is from 86:14 to 70:30.

In one example, the fibrous structures of the present invention comprise one or more trichomes. Non-limiting examples of suitable sources for obtaining trichomes, espe-

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cially trichome fibers, are plants in the Labiatae (Lamiaceae) family commonly referred to as the mint family. Examples of suitable species in the Labiatae family include *Stachys byzantina*, also known as *Stachys lanata* commonly referred to as lamb's ear, woolly betony, or woundwort. The term *Stachys byzantina* as used herein also includes cultivars *Stachys byzantina* 'Primrose Heron', *Stachys byzantina* 'Helene von Stein' (sometimes referred to as *Stachys byzantina* 'Big Ears'), *Stachys byzantina* 'Cotton Boll', *Stachys byzantina* 'Variegated' (sometimes referred to as *Stachys byzantina* 'Striped Phantom'), and *Stachys byzantina* 'Silver Carpet'.

Non-limiting examples of suitable polypropylenes for making the filaments of the present invention are commercially available from Lyondell-Basell and Exxon-Mobil.

Any hydrophobic or non-hydrophilic materials within the fibrous structure, such as polypropylene filaments, may be surface treated and/or melt treated with a hydrophilic modifier. Non-limiting examples of surface treating hydrophilic modifiers include surfactants, such as Triton X-100. Non-limiting examples of melt treating hydrophilic modifiers that are added to the melt, such as the polypropylene melt, prior to spinning filaments, include hydrophilic modifying melt additives such as VW351 and/or S-1416 commercially available from Polyvel, Inc. and Irgasurf commercially available from Ciba. The hydrophilic modifier may be associated with the hydrophobic or non-hydrophilic material at any suitable level known in the art. In one example, the hydrophilic modifier is associated with the hydrophobic or non-hydrophilic material at a level of less than about 20% and/or less than about 15% and/or less than about 10% and/or less than about 5% and/or less than about 3% to about 0% by dry weight of the hydrophobic or non-hydrophilic material.

The fibrous structures of the present invention may include optional additives, each, when present, at individual levels of from about 0% and/or from about 0.01% and/or from about 0.1% and/or from about 1% and/or from about 2% to about 95% and/or to about 80% and/or to about 50% and/or to about 30% and/or to about 20% by dry weight of the fibrous structure. Non-limiting examples of optional additives include permanent wet strength agents, temporary wet strength agents, dry strength agents such as carboxymethylcellulose and/or starch, softening agents, lint reducing agents, opacity increasing agents, wetting agents, odor absorbing agents, perfumes, temperature indicating agents, color agents, dyes, osmotic materials, microbial growth detection agents, antibacterial agents, liquid compositions, surfactants, and mixtures thereof.

The fibrous structure of the present invention may itself be a sanitary tissue product. It may be convolutedly wound about a core to form a roll. It may be combined with one or more other fibrous structures as a ply to form a multi-ply sanitary tissue product. In one example, a co-formed fibrous structure of the present invention may be convolutedly wound about a core to form a roll of co-formed sanitary tissue product. The rolls of sanitary tissue products may also be coreless.

Method for Making a Co-Formed Fibrous Structure

A non-limiting example of a method for making a fibrous structure according to the present invention comprises the steps of: 1) collecting a mixture of filaments and solid additives, such as fibers, for example pulp fibers, onto a collection device, for example a through-air-drying fabric or other fabric or a patterned molding member of the present invention. This step of collecting the filaments and solid additives on the collection device may comprise subjecting

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the co-formed fibrous structure while on the collection device to a consolidation step whereby the co-formed fibrous structure, while present on the collection device, is pressed between a nip, for example a nip formed by a flat or even surface rubber roll and a flat or even surface or patterned, heated (with oil) or unheated metal roll.

In another example, the co-forming method may comprise the steps of a) collecting a plurality of filaments onto a collection device, for example a belt or fabric, such as a patterned molding member, to form a scrim component (a meltblown fibrous structure). The collection of the plurality of filaments onto the collection device to form the scrim component may be vacuum assisted by a vacuum box.

Once the scrim component (meltblown fibrous structure) is formed on the collection device, the next step is to mix, such as commingle, a plurality of solid additives, such as fibers, for example pulp fibers, such as wood pulp fibers, with a plurality of filaments, such as in a coform box, and collecting the mixture on the scrim component carried on the collection device to form a core component. Optionally, an additional scrim component (meltblown fibrous structure) comprising filaments may be added to the core component to sandwich the core component between two scrim components.

The meltblown die used to make the meltblown fibrous structures and/or filaments herein may be a multi-row capillary die and/or a knife-edge die. In one example, the meltblown die is a multi-row capillary die.

NON-LIMITING EXAMPLES

Example 1

A 1.0 gsm meltblown fibrous structure **24** comprising meltblown filaments **23** is laid down upon a collection device **31**, for example an Albany International Velostat170pc740 belt ("forming fabric"), (available from Albany International, Rochester, NH) traveling at 240 ft/min. The meltblown filaments **23** of the meltblown fibrous structure **24** are comprised of 48% LyondellBasell MF650x, 28% LyondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951 and are spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI), at a mass flow of 28 g/min and a ghm of 0.22 and is attenuated with 16.4 kg/min of 204° C. (400° F.) air. An example of this process is shown in FIG. 2B.

Then, fibers **27**, for example pulp fibers such as 440 grams per minute of Koch Industries 4725 semi-treated SSK, are fed into a hammer mill **29** and individualized into fibers **27**, for example cellulose pulp fibers, which are pneumatically conveyed into a coforming box, example of which is described in U.S. Patent Publication No. US 2016/0355950A1 filed Dec. 16, 2015, which is incorporated herein by reference. In the coforming box, the fibers **27**, for example pulp fibers, are commingled with meltblown filaments **23**. The meltblown filaments **23** are comprised of a blend of 48% LyondellBasell MF650x, 28% LyondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951. The meltblown filaments **23** are extruded/spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die, at a ghm of 0.19 and a total mass flow of 93.48 g/min. The meltblown filaments **23** are attenuated with 14 kg/min of about 204° C. (400° F.) air. The mixture (commingled) fibers **27**, for example cellulose pulp fibers and synthetic meltblown filaments **23** are then laid on top of the already formed 1.0 gsm of meltblown fibrous

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structure **24** in the form of a co-formed fibrous structure **22**. An example of this process is shown in FIG. **2B**.

Next, a 1.6 gsm meltblown fibrous structure **24** of the same composition as the meltblown fibrous structure **24** at 0.22 ghm and is attenuated with 16.4 kg/min of 204° C. (400° F.) air is laid down on top of the co-formed fibrous structure **22** such that the co-formed fibrous structure **22** is positioned between the first meltblown fibrous structure **24** and the second meltblown fibrous structure **24** forming a multi-fibrous structure. This multi-fibrous structure is then taken through a nip **33** formed between a steel roll **37** and the forming fabric (collection device **31**), which is backed by a rubber roll **35**, for example a 90 Shore A rubber roll, to form a co-formed fibrous web **28** (co-formed fibrous web ply), an example of which is shown in FIG. **2A**. The steel roll **37** in this example is internally heated with oil to an oil temperature of about 132° C. (270° F.) and is loaded to approximately 90 PLI. The total basis weight of this co-formed fibrous web **28** (co-formed fibrous web ply) is 18.4 gsm. An example of this process is shown in FIG. **2B**.

Two of these co-formed fibrous webs **28** (co-formed fibrous web plies) are then combined on the outside of two paper webs, for example two wet-laid fibrous structures **26** (wet-laid fibrous webs or wet-laid fibrous web plies) of 21 gsm to form an article **20** according to the present invention, as shown in FIG. **4**. The paper webs, for example the wet-laid fibrous structures **26** are pre-formed on a continuous knuckle/discrete pillow patterned molding member with 25% knuckle area. The knuckles of the paper webs, for example the wet-laid fibrous structures are facing out relative to the article **20**, as are the 1.6 gsm meltblown fibrous structures **24** (scrims), when present, relative to the article **20**. In other words, when present, the meltblown fibrous structures **24** form at least one exterior surface of the article **20**. The four fibrous webs (fibrous web plies) (co-formed fibrous web ply/wet-laid fibrous web ply/wet-laid fibrous web ply/co-formed fibrous web ply) are then bonded together at 60 feet per minute in a pin-pin steel thermal bond unit, oil heated to about 143° C. (290° F.) and loaded to 200 psi of pressure on two 2.5" diameter cylinders.

Each of the 21 gsm paper webs, for example wet-laid fibrous structures **26** are formed on an AstenJohnson 866A forming wire (AstenJohnson, Charleston, SC), then vacuum transferred to the patterned molding member described above. A pulp blend of 40% lightly refined GPOP NSK pulp (Georgia-Pacific Corporation, Atlanta, GA), 20% Alabama River southern softwood kraft (Georgia-Pacific Corporation, Atlanta, GA), and 40% *eucalyptus* pulp (Fibria Celulose S.A., Sao Paulo, Brazil). Wet-end additives include 10 #/ton Kymene, 2 #/ton Finnfix CMC and 1 #/T Wickit 1285 surfactant (all commercially available). The papermachine is run at 750 fpm Yankee speed in through-air-dry (TAD) mode, with 2% wet micro-contraction and 18% crepe. The wet-laid fibrous structure is creped from the Yankee with a 25 degree bevel creping blade and 81 degree impact angle. The wet-laid fibrous structure is then wound up on a papermachine reel that is run at 615 fpm to form a parent roll of a wet-laid fibrous web (wet-laid fibrous web ply). The parent roll is then unwound during the article making process.

Example 2

An approximately 1.0 gsm meltblown fibrous structure **24** is laid down upon a collection device **31**, for example an Albany International Velostat170pc740 belt ("forming fabric") (available from Albany International, Rochester, NH)

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traveling at 240 ft/min. The meltblown filaments **23** of the meltblown fibrous structure **24** are comprised of 48% LynondellBasell MF650x, 28% LynondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951 and are spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI), at a mass flow of 28 g/min and a ghm of 0.22 and is attenuated with 16.4 kg/min of 204° C. (400° F.) air. An example of this process is shown in FIG. **2B**.

Then, fibers **27**, for example pulp fibers such as 440 grams per minute of Resolute CoosAbsorb ST semi-treated SSK (Resolut Forest Products, Montreal, Quebec, Canada), are fed into a hammer mill **29** and individualized into fibers **27**, for example cellulose pulp fibers, which are pneumatically conveyed into a coforming box like Example 1 above. In the coforming box, the fibers **27**, for example pulp fibers are commingled with meltblown filaments **23**. The meltblown filaments **23** are comprised of a blend of 48% Lynondell-Basell MF650x, 28% LynondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951. The meltblown filaments **23** are extruded/spun from a die **25**, for example a multi-row capillary die at a ghm of 0.19 and a total mass flow of 93.48 g/min like Example 1 above. The meltblown filaments **23** are attenuated with 14 kg/min of 204° C. (400° F.) air. The mixture (commingled) fibers **27**, for example cellulose pulp fibers and synthetic meltblown filaments **23** are then laid on top of the already formed 1.0 gsm of meltblown fibrous structure **24** in the form of a co-formed fibrous structure **22**. An example of this process is shown in FIG. **2B**.

Next, a 1.6 gsm meltblown fibrous structure **24** of the same composition as the meltblown fibrous structure **24** at 0.22 ghm and is attenuated with 16.4 kg/min of 204° C. (400° F.) air is laid down on top of the co-formed fibrous structure **22** such that the co-formed fibrous structure **22** is positioned between the first meltblown fibrous structure **24** and the second meltblown fibrous structure **24** to form a multi-fibrous structure. This multi-fibrous structure is then taken through a nip **33** formed between a steel roll **37** and the forming fabric (collection device **31**), which is backed by a rubber roll **35**, for example a 90 Shore A rubber roll, to form a co-formed fibrous web **28** (co-formed fibrous web ply), an example of which is shown in FIG. **2A**. The steel roll **37** in this example is internally heated with oil to an oil temperature of about 132° C. (270° F.) and is loaded to approximately 90 PLI. The total basis weight of this co-formed fibrous web **28** (co-formed fibrous web ply) is 18.4 gsm. An example of this process is shown in FIG. **2B**.

Two of these co-formed fibrous webs **28** (co-formed fibrous web plies) are then combined on the outside of two paper webs, for example two wet-laid fibrous structures **26** (wet-laid fibrous webs or wet-laid fibrous web plies) of 21 gsm to form an article **20** according to the present invention, as shown in FIG. **4**. The paper webs, for example wet-laid fibrous structures **26** are pre-formed on a continuous knuckle/discrete pillow patterned molding member with 45% knuckle area. The knuckles of the paper webs, for example wet-laid fibrous structures **26** are facing out relative to the article **20**, as are the 1.6 gsm meltblown fibrous structures **24** (scrims), when present, relative to the article **20**, such that at least one of the meltblown fibrous structures **24** forms an exterior surface of the article **20** when present. The four fibrous webs (fibrous web plies) (co-formed fibrous web ply/wet-laid fibrous web ply/wet-laid fibrous web ply/co-formed fibrous web ply) are then bonded together at 60 feet per minute in a pin-pin steel thermal bond unit, oil

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heated to about 140° C. (285° F.) and loaded to 150 psi of pressure on two 2.5" diameter cylinders.

Each of the 21 gsm paper webs, for example wet-laid fibrous structures **26** is formed on an AstenJohnson 866A forming wire (AstenJohnson, Charleston, SC), then vacuum transferred to the patterned molding member described above. A pulp blend of 40% lightly refined GPOP NSK pulp (Georgia-Pacific Corporation, Atlanta, GA), 20% Alabama River southern softwood kraft (Georgia-Pacific Corporation, Atlanta, GA), and 40% *eucalyptus* pulp (Fibria Celulose S.A., Sao Paulo, Brazil). Wet-end additives include 10 #/ton Kymene, 2 #/ton Finfix CMC and 1 #/T Wickit 1285 surfactant (all commercially available). The papermachine is run at 700 fpm Yankee speed in through-air-dry (TAD) mode, with 2% wet micro-contraction and 18% crepe. The wet-laid fibrous structure is creped from the Yankee with a 25 degree bevel creping blade and 81 degree impact angle. The wet-laid fibrous structure is then wound up on a papermachine reel that is run at 574 fpm (feet per minute) to form a parent roll of a wet-laid fibrous web (wet-laid fibrous web ply). The parent roll is then unwound during the article making process.

Example 3

A 28.2 gsm paper web, for example wet-laid fibrous structure **26** or wet-laid fibrous web (wet-laid fibrous web ply) made on a continuous knuckle/discrete pillow patterned molding member with 25% knuckle area is unwound upon an Albany International Velostat 170pc740 belt (Albany International) traveling at 155 fpm. Laid upon this paper web, for example wet-laid fibrous structure **26** is 2.0 gsm of a meltblown fibrous structure **24** comprising meltblown filaments **23** comprised of 48% LyondellBasell MF650x, 28% LyondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951. The meltblown filaments **23** are extruded/spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI), at a ghm of 0.19 and a total mass flow of 93.48 g/min like Example 1 above. The meltblown filaments **23** are attenuated with 14 kg/min of 204° C. (400° F.) air. In this example this is now ply A.

An approximately 1.1 gsm meltblown fibrous structure **24** is laid down upon a collection device **31**, for example an Albany International Velostat170pc740 belt ("forming fabric") (available from Albany International, Rochester, NH) traveling at 220 ft/min. The meltblown filaments **23** of the meltblown fibrous structure **24** are comprised of 48% LyondellBasell MF650x, 28% LyondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951 and are spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI) at a mass flow of 28 g/min and a ghm of 0.22 and is attenuated with 16.4 kg/min of 204° C. (400° F.) air. An example of this process is shown in FIG. 2B.

Then, fibers **27**, for example pulp fibers such as 400 grams per minute of Resolute CoosAbsorb ST semi-treated SSK (Resolut Forest Products, Montreal, Quebec, Canada), are fed into a hammer mill **29** and individualized into fibers **27**, for example cellulose pulp fibers, which are pneumatically conveyed into a coforming box like Example 1 above. In the coforming box, the fibers **27**, for example pulp fibers are commingled with meltblown filaments **23**. The meltblown filaments **23** are comprised of a blend of 48% LyondellBasell MF650x, 28% LyondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet

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412951. The meltblown filaments **23** are extruded/spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI) at a ghm of 0.19 and a total mass flow of 93.48 g/min like Example 1 above. The meltblown filaments **23** are attenuated with 14 kg/min of 204° C. (400° F.) air. The mixture (commingled) fibers **27**, for example cellulose pulp fibers and synthetic meltblown filaments **23** are then laid on top of the already formed 1.1 gsm of meltblown fibrous structure **24** in the form of a co-formed fibrous structure **22**. An example of this process is shown in FIG. 2B.

Next, a 1.6 gsm meltblown fibrous structure **24** of the same composition as the meltblown fibrous structure **24** at 0.22 ghm and is attenuated with 16.4 kg/min of 204° C. (400° F.) air is laid down on top of the co-formed fibrous structure **22** such that the co-formed fibrous structure **22** is positioned between the first meltblown fibrous structure **24** and the second meltblown fibrous structure **24** to form a multi-fibrous structure. This multi-fibrous structure is then taken through a nip **33** formed between a steel roll **37** and the forming fabric (collection device **31**), which is backed by a rubber roll **35**, for example a 90 Shore A rubber roll, to form a co-formed fibrous web **28** (co-formed fibrous web ply), an example of which is shown in FIG. 2A. The steel roll **37** in this example is internally heated with oil to an oil temperature of about 132° C. (270° F.) and is loaded to approximately 90 PLI. The total basis weight of this co-formed fibrous web **28** (co-formed fibrous web ply) is 19.4 gsm. An example of this process is shown in FIG. 2B. This is ply B in this example.

In a separate process, two ply A paper webs, for example wet-laid fibrous structures **26** and/or wet-laid fibrous webs are combined with a ply B co-formed fibrous web **28** to form an article **20** as shown in FIG. 18. The ply A paper webs, for example wet-laid fibrous structures **26** and/or wet-laid fibrous webs, are combined with the meltblown filaments **24** facing the outside of the article **20**. These plies are then bonded together at 60 feet per minute in a pin-pin steel thermal bond unit, oil heated to about 140° C. (285° F.) and loaded to 150 psi pressure on two 2.5" diameter cylinders.

The 28.2 gsm paper web, for example wet-laid fibrous structure **26** and/or wet-laid fibrous web (wet-laid fibrous web ply) is formed on an AstenJohnson 866A forming wire (AstenJohnson) like above, then vacuum transferred to a continuous knuckle/discrete pillow patterned molding member with 25% knuckle area. A pulp fiber blend of 40% refined (to 15 PFR) GPOP NSK pulp (Georgia-Pacific Corporation), 30% West Fraser CTMP (West Fraser, Vancouver, British Columbia, Canada), and 30% *eucalyptus* pulp (Fibria Celulose S.A.) is used. Wet-end additives include 15 #/ton Kymene, 4.5 #/ton Finfix CMC and 1 #/T Wickit 1285 surfactant (all commercially available). The papermachine is run at 600 fpm in through-air-dry (TAD) mode, with 10% wet micro-contraction and 10% crepe. The wet-laid fibrous structure is creped from the Yankee with a 25 degree bevel creping blade and 81 degree impact angle. The wet-laid fibrous structure is then wound up on a papermachine reel that is run at 555 fpm (feet per minute) to form a parent roll of a wet-laid fibrous web (wet-laid fibrous web ply). The parent roll is then unwound during the article making process.

Example 4

An approximately 1.1 gsm meltblown fibrous structure **24** is laid down upon a collection device **31**, for example an Albany International Velostat170pc740 belt ("forming fab-

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ric") (available from Albany International, Rochester, NH) traveling at 215 ft/min (fpm). The meltblown filaments **23** of the meltblown fibrous structure **24** are comprised of 48% LyondellBasell MF650x, 28% LyondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951 and are spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI) at a mass flow of 28 g/min and a ghm of 0.22 and is attenuated with 16.4 kg/min of 204° C. (400° F.) air. An example of this process is shown in FIG. 2B.

Then, fibers **27**, for example pulp fibers such as 495 grams per minute of Resolute CoosAbsorb ST semi-treated SSK (Resolut Forest Products, Montreal, Quebec, Canada) are fed into a hammer mill **29** and individualized into fibers **27**, for example cellulose pulp fibers, which are pneumatically conveyed into a coforming box like Example 1 above. In the coforming box, the fibers **27**, for example pulp fibers are commingled with meltblown filaments **23**. The meltblown filaments **23** are comprised of a blend of 48% LyondellBasell MF650x, 28% LyondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951. The meltblown filaments **23** are extruded/spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI), at a ghm of 0.19 and a total mass flow of 93.48 g/min like Example 1 above. The meltblown filaments **23** are attenuated with 14 kg/min of 204° C. (400° F.) air. The mixture (commingled) fibers **27**, for example cellulose pulp fibers and synthetic meltblown filaments **23** are then laid on top of the already formed 1.1 gsm of meltblown fibrous structure **24** in the form of a co-formed fibrous structure **22**.

Next, a 1.6 gsm meltblown fibrous structure **24** of the same composition as the meltblown fibrous structure **24** at 0.22 ghm and is attenuated with 16.4 kg/min of 204° C. (400° F.) air is laid down on top of the co-formed fibrous structure **22** such that the co-formed fibrous structure **22** is positioned between the first meltblown fibrous structure **24** and the second meltblown fibrous structure **24** forming a multi-fibrous structure, a co-formed fibrous web **28**. The total basis weight of this co-formed fibrous web **28** is 23.4 gsm. An example of this process is shown in FIG. 2B. This is now ply A in this example.

In a separate process, one ply A co-formed fibrous web **28** is combined between two 28.2 gsm paper webs, for example two wet-laid fibrous structures **26** or wet-laid fibrous webs (wet-laid fibrous web plies). These paper webs, for example wet-laid fibrous structures **26** and/or wet-laid fibrous webs are formed on a continuous knuckle molding member and are combined with the continuous pillow pattern facing outwards. These plies and/or fibrous structures and/or webs are then bonded together at 60 feet per minute in a pin-pin steel thermal bonding unit which is oil heated to an oil temp of about 160° C. (320° F.) and loaded to 200 psi of pressure on two 2.5" diameter cylinders.

The 28.2 gsm paper web, for example wet-laid fibrous structure **26** or wet-laid fibrous web (wet-laid fibrous web ply) is formed on an AstenJohnson 866A forming wire (AstenJohnson) like above, then vacuum transferred to a continuous pillow/discrete knuckle patterned molding member. A pulp fiber blend of 40% refined (to 15 PFR) GPOP NSK pulp (Georgia-Pacific Corporation), 30% West Fraser CTMP (West Fraser, Vancouver, British Columbia, Canada), and 30% *eucalyptus* pulp (Fibria Celulose S.A.) is used. Wet-end additives include 15 #/ton Kymene, 4.5 #/ton Finnfix CMC and 1 #/T Wickit 1285 surfactant (all commercially available). The papermachine is run at 700 fpm in

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through-air-dry (TAD) mode, with 15% wet micro-contraction and +5% crepe (reel faster than Yankee). The wet-laid fibrous structure is creped from the Yankee with a 45 degree bevel creping blade and 101 degree impact angle. The wet-laid fibrous structure is then wound up on a papermachine reel that is run at 735 fpm (feet per minute) to form a parent roll of a wet-laid fibrous web (wet-laid fibrous web ply). The parent roll is then unwound during the article making process.

Example 5

A 23.1 gsm paper web, for example a wet-laid fibrous structure **26** or wet-laid fibrous web (wet-laid fibrous web ply) which is made on a continuous knuckle/discrete pillow molding member with a 25% knuckle area is unwound onto a patterned molding member, knuckles facing away from the patterned molding member, traveling at 220 ft/minute.

Next, an approximately 1.1 gsm meltblown fibrous structure **24** is laid down upon the paper web, for example wet-laid fibrous structure **26** and/or wet-laid fibrous web. The meltblown filaments **23** of the meltblown fibrous structure **24** are comprised of 48% LyondellBasell MF650x, 28% LyondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951 and are spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI) at a mass flow of 28 g/min and a ghm of 0.22 and is attenuated with 16.4 kg/min of 204° C. (400° F.) air. An example of this process is shown in FIG. 2B.

Then, fibers **27**, for example pulp fibers such as 325 grams per minute of Resolute CoosAbsorb ST semi-treated SSK (Resolut Forest Products, Montreal, Quebec, Canada) are fed into a hammer mill **29** and individualized into fibers **27**, for example cellulose pulp fibers, which are pneumatically conveyed into a coforming box like Example 1 above. In the coforming box, the fibers **27**, for example pulp fibers are commingled with meltblown filaments **23**. The meltblown filaments **23** are comprised of a blend of 48% LyondellBasell MF650x, 28% LyondellBasell MF650w, 17% LyondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951. The meltblown filaments **23** are extruded/spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI) at a ghm of 0.19 and a total mass flow of 93.48 g/min like Example 1 above. The meltblown filaments **23** are attenuated with 14 kg/min of 204° C. (400° F.) air. The mixture (commingled) fibers **27**, for example cellulose pulp fibers and synthetic meltblown filaments **23** are then laid on top of the already formed 23.1 gsm paper web, for example wet-laid fibrous structure **26** and/or wet-laid fibrous web, which has its knuckles facing outward in the form of a co-formed fibrous structure **22**.

Next, a 1.6 gsm meltblown fibrous structure **24** of the same composition at a ghm of 0.22 and attenuated with 16.4 kg/min of 204° C. (400° F.) air is laid down on top of the co-formed fibrous structure **22** to form a multi-fibrous structure. This multi-fibrous structure is then taken through a nip **33** formed between a steel roll **37** and the forming fabric (collection device **31**), which is backed by a rubber roll **35**, for example a 90 Shore A rubber roll. The steel roll **37** in this example is internally heated with oil to an oil temperature of about 132° C. (270° F.) and is loaded to approximately 90 PLI. The total weight of this web is about 40.1 gsm. In this example this is now ply A.

Then a 2.0 gsm meltblown fibrous structure **24** of the same composition, ghm, and attenuation air settings as

described immediately above is applied to the surface of the paper web, for example wet-laid fibrous structure **26** of ply A. This multi-fibrous structure is now 42.1 gsm and is referred to as ply B in this example.

In a separate process, two ply B paper webs, for example two wet-laid fibrous structures **26** and/or wet-laid fibrous webs are combined with the paper webs, for example wet-laid fibrous structures **26** and/or wet-laid fibrous webs facing inward to form an article **20** as shown in FIGS. **22A** and **22B**. These plies, fibrous structures and/or web are then bonded together at 60 feet per minute in a pin-pin steel thermal bonding unit which is oil heated to an oil temp of about 143° C. (290° F.) and loaded to 200 psi of pressure on two 2.5" diameter cylinders. An example of this process is shown in FIG. **23B**.

The 23.1 gsm paper web, for example wet-laid fibrous structure **26** and/or wet-laid fibrous web (wet-laid fibrous web ply) is formed on an AstenJohnson 866A forming wire (AstenJohnson), then vacuum transferred to a continuous knuckle/discrete pillow patterned molding member with 25% knuckle area. A pulp fiber blend of 40% unrefined GPOP NSK pulp (Georgia-Pacific Corporation), 20% West Fraser CTMP (West Fraser, Vancouver, British Columbia, Canada), and 40% *eucalyptus* pulp (Fibria Celulose S.A.) is used. Wet-end additives include 15 #/ton Kymene, 4.5 #/ton Finnfix CMC and 1 #/T Wickit 1285 surfactant (all commercially available). The papermachine is run at 700 fpm in through-air-dry (TAD) mode, with 2% wet micro-contraction and 18% crepe. The wet-laid fibrous structure is creped from the Yankee with a 25 degree bevel creping blade and 81 degree impact angle. The wet-laid fibrous structure is then wound up on a papermachine reel that is run at 574 fpm (feet per minute) to form a parent roll of a wet-laid fibrous web (wet-laid fibrous web ply). The parent roll is then unwound during the article making process.

Example 6

A 23.1 gsm paper web, for example a wet-laid fibrous structure **26** and/or wet-laid fibrous web (wet-laid fibrous web ply) which is made on a continuous knuckle/discrete pillow molding member with a 25% knuckle area is unwound onto a patterned molding member, knuckles facing away from the patterned molding member, traveling at 220 ft/minute.

Then, fibers **27**, for example pulp fibers such as 325 grams per minute of Resolute CoosAbsorb ST semi-treated SSK (Resolut Forest Products, Montreal, Quebec, Canada) are fed into a hammer mill **29** and individualized into fibers **27**, for example cellulose pulp fibers, which are pneumatically conveyed into a coforming box like Example 1 above. In the coforming box, the fibers **27**, for example pulp fibers are commingled with meltblown filaments **23**. The meltblown filaments **23** are comprised of a blend of 48% Lynondell-Basell MF650x, 28% LynondellBasell MF650w, 17% LynondellBasell PH835, 5% Polyvel S1416, and 2% Ampacet 412951. The meltblown filaments **23** are extruded/spun from a die **25**, for example a multi-row capillary Biax-Fiberfilm die (Biax-Fiberfilm Corporation, Greenville, WI) at a ghm of 0.19 and a total mass flow of 93.48 g/min like Example 1 above. The meltblown filaments **23** are attenuated with 14 kg/min of 204° C. (400° F.) air. The mixture (commingled) fibers **27**, for example cellulose pulp fibers and synthetic meltblown filaments **23** are then laid on top of the already formed 23.1 gsm paper web, for example wet-laid fibrous

structure **26** and/or wet-laid fibrous web, which has its knuckles facing outward in the form of a co-formed fibrous structure **22**.

Next, a 1.6 gsm meltblown fibrous structure **24** of the same composition at a ghm of 0.22 and attenuated with 16.4 kg/min of 204° C. (400° F.) air is laid down on top of the co-formed fibrous structure **22** forming a multi-fibrous structure. This multi-fibrous structure is then taken through a nip **33** formed between a steel roll **37** and the forming fabric (collection device **31**), which is backed by a rubber roll **35**, for example a 90 Shore A rubber roll. The steel roll **37** in this example is internally heated with oil to an oil temperature of about 132° C. (270° F.) and is loaded to approximately 90 PLI. The total basis weight of this combined multi-fibrous structure and/or multi-fibrous web is 39 gsm. This is now ply A in this example.

Then a 2.0 gsm meltblown fibrous structure **24** of the same composition, ghm, and attenuation air settings as described immediately above is applied to the surface of the paper web, for example wet-laid fibrous structure **26** of ply A. This multi-fibrous structure is now 41 gsm and is referred to as ply B in this example.

In a separate process, one ply A is combined with one ply B. These plies are then bonded together at 60 feet per minute in a pin-pin steel thermal bonding unit which is oil heated to an oil temp of about 143° C. (290° F.) and loaded to 200 psi of pressure on two 2.5" diameter cylinders.

The 23.1 gsm paper web, for example wet-laid fibrous structure **26** or wet-laid fibrous web (wet-laid fibrous web ply) is formed on an AstenJohnson 866A forming wire (AstenJohnson), then vacuum transferred to a continuous knuckle/discrete pillow patterned molding member with 25% knuckle area. A pulp fiber blend of 40% unrefined GPOP NSK pulp (Georgia-Pacific Corporation), 20% West Fraser CTMP (West Fraser, Vancouver, British Columbia, Canada), and 40% *eucalyptus* pulp (Fibria Celulose S.A.) is used. Wet-end additives include 15 #/ton Kymene, 4.5 #/ton Finnfix CMC and 1 #/T Wickit 1285 surfactant (all commercially available). The papermachine is run at 700 fpm in through-air-dry (TAD) mode, with 2% wet micro-contraction and 18% crepe. The wet-laid fibrous structure is creped from the Yankee with a 25 degree bevel creping blade and 81 degree impact angle. The wet-laid fibrous structure is then wound up on a papermachine reel that is run at 574 fpm (feet per minute) to form a parent roll of a wet-laid fibrous web (wet-laid fibrous web ply). The parent roll is then unwound during the article making process.

Test Methods

Unless otherwise specified, all tests described herein including those described under the Definitions section and the following test methods are conducted on samples that have been conditioned in a conditioned room at a temperature of 23° C. ±1.0° C. and a relative humidity of 50% ±2% for a minimum of 24 hours prior to the test. These will be considered standard conditioning temperature and humidity. All plastic and paper board packaging articles of manufacture, if any, must be carefully removed from the samples prior to testing. The samples tested are "usable units." "Usable units" as used herein means sheets, flats from roll stock, pre-converted flats, fibrous structure, and/or single or multi-ply products. Except where noted all tests are conducted in such conditioned room, under the same environmental conditions in such conditioned room. Discard any damaged product. Do not test samples that have defects such as wrinkles, tears, holes, and like. All instruments are

calibrated according to manufacturer's specifications. The stated number of replicate samples to be tested is the minimum number.

Basis Weight Test Method

Basis weight of an article and/or fibrous web and/or fibrous structure is measured on stacks of eight to twelve usable units using a top loading analytical balance with a resolution of ± 0.001 g. A precision cutting die, measuring 8.890 cm by 8.890 cm or 10.16 cm by 10.16 cm is used to prepare all samples.

Condition samples under the standard conditioning temperature and humidity for a minimum of 10 minutes prior to cutting the sample. With a precision cutting die, cut the samples into squares. Combine the cut squares to form a stack eight to twelve samples thick. Measure the mass of the sample stack and record the result to the nearest 0.001 g. Calculations:

$$\text{Basis Weight, g/m}^2 = \frac{\text{mass of stack}}{(\text{area of 1 square in stack})(\# \text{ squares in stack})}$$

Report result to the nearest 0.1 g/m². Sample dimensions can be changed or varied using a similar precision cutter as mentioned above, so as at least 645 square centimeters of sample area is in the stack.

Individual fibrous structures and/or fibrous webs that are ultimately combined to form an article may be collected during their respective making operation prior to combining with other fibrous web and/or fibrous structures and then the basis weight of the respective fibrous web and/or fibrous structure is measured as outlined above.

Average Diameter Test Method

There are many ways to measure the diameter of a fiber. One way is by optical measurement. An article and/or fibrous web and/or fibrous structure comprising filaments is cut into a rectangular shape sample, approximately 20 mm by 35 mm. The sample is then coated using a SEM sputter coater (EMS Inc, PA, USA) with gold so as to make the filaments relatively opaque. Typical coating thickness is between 50 and 250 nm. The sample is then mounted between two standard microscope slides and compressed together using small binder clips. The sample is imaged using a 10× objective on an Olympus BHS microscope with the microscope light-collimating lens moved as far from the objective lens as possible. Images are captured using a Nikon D1 digital camera. A Glass microscope micrometer is used to calibrate the spatial distances of the images. The approximate resolution of the images is 1 μm/pixel. Images will typically show a distinct bimodal distribution in the intensity histogram corresponding to the filaments and the background. Camera adjustments or different basis weights are used to achieve an acceptable bimodal distribution. Typically 10 images per sample are taken and the image analysis results averaged.

The images are analyzed in a similar manner to that described by B. Pourdeyhimi, R. and R. Dent in "Measuring fiber diameter distribution in nonwovens" (Textile Res. J. 69(4) 233-236, 1999). Digital images are analyzed by computer using the MATLAB (Version. 6.1) and the MATLAB Image Processing Tool Box (Version 3.) The image is first converted into a grayscale. The image is then binarized into black and white pixels using a threshold value that minimizes the intraclass variance of the thresholded black and white pixels. Once the image has been binarized, the image is skeletonized to locate the center of each fiber in the image.

The distance transform of the binarized image is also computed. The scalar product of the skeletonized image and the distance map provides an image whose pixel intensity is either zero or the radius of the fiber at that location. Pixels within one radius of the junction between two overlapping fibers are not counted if the distance they represent is smaller than the radius of the junction. The remaining pixels are then used to compute a length-weighted histogram of filament diameters contained in the image.

The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as "40 mm" is intended to mean "about 40 mm".

Every document cited herein, including any cross referenced or related patent or application and any patent application or patent to which this application claims priority or benefit thereof, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

What is claimed is:

1. An article comprising:

- a. a first ply comprising a first plurality of wood pulp fibers and a plurality of spun polyvinyl alcohol filaments, wherein a first portion of the first plurality of wood pulp fibers are in the form of a first wet laid fibrous structure and a second portion of the first plurality of wood pulp fibers are commingled together with at least a portion of the plurality of spun polyvinyl alcohol filaments in the form of a coform fibrous structure, wherein the portion of the plurality of spun polyvinyl alcohol filaments;
- b. a second ply comprising a second wet laid fibrous structure comprising a second plurality of wood pulp fibers, wherein the second wet laid fibrous structure is bonded to the first wet laid fibrous structure; and wherein the article is void of a hydroentangled fibrous web.

2. The article according to claim 1 wherein at least one the first and second plurality of wood pulp fibers are selected from the group consisting of: northern softwood kraft pulp fibers, southern softwood kraft pulp fibers, northern hardwood pulp fibers, tropical hardwood pulp fibers, and mixtures thereof.

3. The article according to claim 1 wherein at least one of the first plurality of wood pulp fibers or the second plurality of wood pulp fibers comprise trichome fiber.

4. The article according to claim 1 wherein at least one of the first plurality of wood pulp fibers or the second plurality of wood pulp fibers comprises an absorbent gel material.

5. The article according to claim 1 wherein at least one of the first plurality of wood pulp fibers or the second plurality of wood pulp fibers comprises a surface having a surface pattern.

6. The article according to claim 1 wherein the article 5 comprises at least one meltblown fibrous structure that forms at least one of the exterior surfaces of the article.

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